WINCOR NIXDORF

RBC Romanian Business Consult

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1. Romanian New Fiscal EJ Law:

Romania is a fiscal country.
To cash-up money it is mandatory to use a solution homologated by the Romanian fiscal authorities. The homologation / fiscal certification will occur on the following elements:

- The Fiscal Printer including Fiscal Controller, Electronic Journal ("EJ") and Data Communication Module ("CoMo");
- The Printer Customer Display ("CD");
- The Fiscal POS application

During the certification process it will be certified only the fiscal POS application together with the fiscal printer. The Back Store and Back Office software will not be part of this process. Sometimes the certification committee will check some values in back store/back office to the sure that the data was correctly synchronized and processed.

If the Customer Display, Printer, Electronic Journal, Transmission Module or the Fiscal controllers is out of service, the POS solution has to stop sales. It's therefore forbidden to cash up.

The aim of the homologation / fiscal certification is to enable the Romanian Fiscal Authority (ANAF) to know the exact turnover performed by the retailers and therefore to know the corresponding VAT values (per VAT rates) that has to be collected. For these reasons in the certification procedure, will be tested the fiscal receipts, fiscal reports (including reports from Electronic Journal) and also will be tested Data Transmission to ANAF server.

2. Currency:

The currency in Romania is the LEU (used in retail is "Lei" as plural). It's a currency with 2 decimals.
The equivalent for "Cent" is called "BAN"
The symbol used for this currency is "RON" or "LEI" but usually is used "Lei".
The smallest bank note is 1 Lei (value = 0,22 EUR approximatively)

Some western countries do not work with 1 cent (coin), and the lowest denomination is 5 cents (coin). From this reason, if the total of the receipt is 17,43 Euro, the Cash Register work in two ways:
Rounding the total to 17,45
Or
Rounding the change to the value divided by 5 cents.
In Romania it is not allowed to round any value (total or change), even if the cashier do not return the correct change to client.

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3. Fiscal Printer:

FEED

If you push this button once and release it, the printer feeds paper for one line (1/6 inch).
If you push this button and hold it down, the printer feeds the paper as long as the button is not released.
The button can be locked by the application software and then will be without function.

POWER

All LED off: power is not stable
Green POWER LED on: power is stable
Green POWER LED blinking: printing speed may be low (*) if necessary contact your technical support

PAPER

Yellow PAPER LED off: Paper is properly inserted. Yellow PAPER LED on: Paper roll near end.
Yellow PAPER and red ERROR on: Paper end is reached.

ERROR

Red ERROR LED off: normal condition
Red ERROR LED on: not ready for operating. Printer cover is not closed or in combination with Yellow PAPER LED on, paper end is reached
Red ERROR LED blinking: an error occurs. Switch the printer off and on again. In case of no success contact your technical support.

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LED overview (*additional MF-LED's see page*)

	POWER green	PAPER yellow	ERROR red	Meaning
	off	**off**	**off**	No power
Operation	**on**			Power on
	blinking			If necessary call for technical support
		off		Paper properly inserted
Paper		**on**		Paper near end
		on	**on**	Paper end
Error			**blinking**	If necessary call for technical support
			on	Cover not closed

3.1 MF-TH230+ RO: MF-related Features

The MF-TH230+ RO has some special features due to fiscal law requirements.
According to the fiscal legislation the MF-TH230+ RO is sealed from outside. It incorporates extra hardware (fiscal interface board, fiscal memory, EJ SD card and communication module) to comply fully with all country-specific fiscal requirements.

Fiscal Memory Fixed to the printer chassis under separate seal; storing all VAT-related data (daily turnovers)

SD Card Located under the printer cover sealed; storing all Electronic Journal data.

Sealing pot Before installation of the MF-TH230+ RO the sealing pot is to be sealed according to the country-specific procedure.

MF LED's The MF LED's are located on the right side of the printer. They are controlled by the fiscal firmware. The specification document shows the possible different blink codes and their meaning.

3.2 Self-test printout

1) Turn off the printer. Press the On/Off button for at least one second.

2) Push the FEED button while turning on the printer. A service menu will be printed.

3) Press the FEED button two times shortly to select the printer self-test (NOT Fiscal printer self-test). Press same button for more than one second

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TH230+ Printer
Service Menu

1 Exit/Save
2 Print Self Test
3 Diagnostic
4 Configuration
5 Information

Press the FEED button short to scroll through the menu and longer than 1 sec to enter the topic!

TH230+
Self Test

Hardware Information:

Manufacturer	: WINCOR-NIXDORF
Serial Number	: 97AX642807
Manufactured Board	: K
Production Date	: 14-DEC-2016
Interface ID	: 05
Interface Board	: RS232 Cash Drawer Customer Display MF-WN PS/2

Software Modules:

Firmware	: 01.10
Booter	: 01.01
Loader	: 01.04
Status	: Verified
Power Up Test	: 01.00
Checksum	: OK
Character Font	: StdCodePages
Font Version	: 01.04

Printer Settings:

Black Mark Sensor	: disabled
Endurance Test	: disabled
Automatic LF	: enabled
Power Button	: enabled
Power-ON Notice	: disabled
Paper-NE Notice	: enabled
Receipt Shooting	: disabled
Shoot Flush Time	: 63.75 sec
Page Mode	: enabled
Sleep Mode	: disabled
Max. Power	: 60 W
Max. Speed	: 220 mm/sec
Endurance Tickets	: 500
Paper Width	: 80 mm
Print Density	: 100 %
Paper Type	: single color
Code-128 ChkDigit	: enabled
ITF Leading Zero	: enabled
Brc String Term	: enabled
Leg. Printer Supp.	: disabled
Listen to RT-Cmds	: always
Default Code Page	: 0
Autorecovery	: Off
Orig.-Controller	: yes

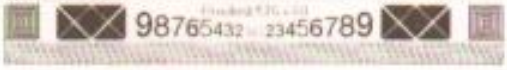
Sensor Information:

Sensor Plug Detection	
Black Mark Sensor	: not available

Measured Values:

Supply Voltage	: 25.2 V
Printhead Temp	: 26.8 °C

Pixel Test:



*** completed ***

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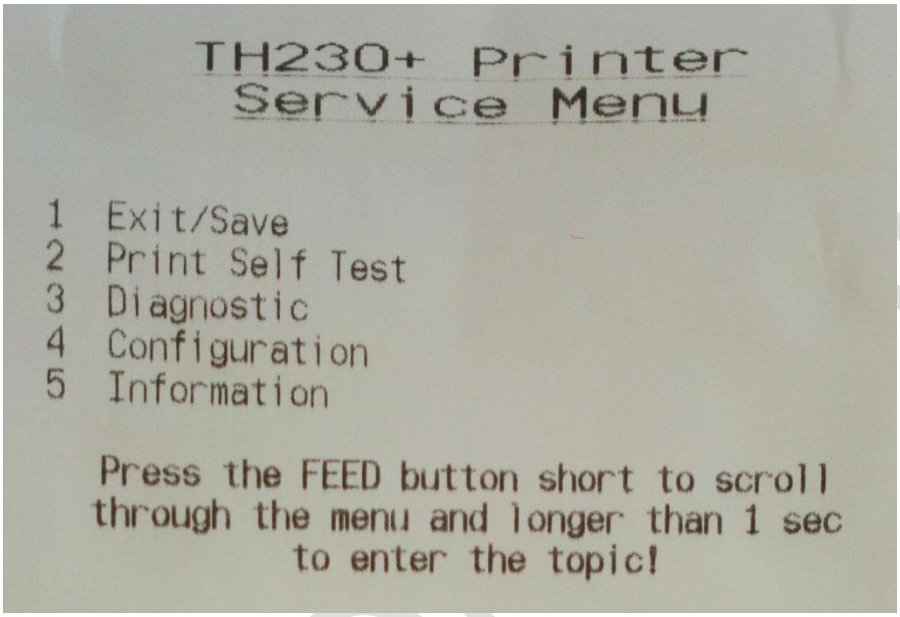
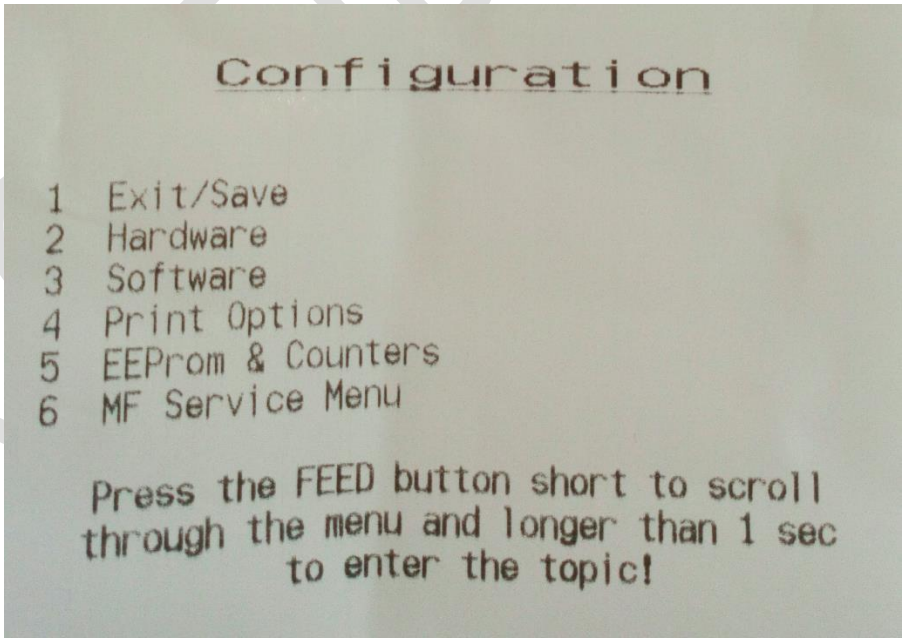
4. Procedure to obtain "Print Fiscal SelfTest"

a. The printer must be turned off.
b. Press the FEED and Power buttons at the same time to power on the printer. The printer will enter in the Service Menu and print the fallowing message:

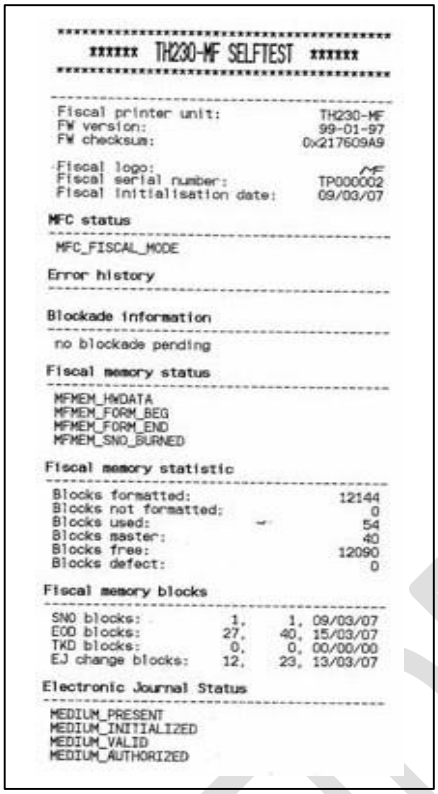


c. Press the FEED button 4 times short and 1 time long, the printer will enter in the Configuration menu and print the fallowing message:



d. Press the FEED button 6 times short and 1 time long, the printer will enter the MF Service Menu

e. Press the FEED button 1 time short and 1 time long, the FISCAL SELFTEST report will be printed.



```
****************************************
****** TH230-MF SELFTEST ******
****************************************

Fiscal printer unit:            TH230-MF
FW version:                     99-01-97
FW checksum:                  0x217609A9

Fiscal logo:                          MF
Fiscal serial number:           TP000002
Fiscal initialisation date:     09/03/07

MFC status
MFC_FISCAL_MODE

Error history

Blockade information
no blockade pending

Fiscal memory status
MFMEM_HWDATA
MFMEM_FORM_REG
MFMEM_FORM_END
MFMEM_SNO_BURNED

Fiscal memory statistic
Blocks formatted:                12144
Blocks not formatted:                0
Blocks used:                        54
Blocks master:                      40
Blocks free:                     12090
Blocks defect:                       0

Fiscal memory blocks
SNO blocks:        1,    1, 09/03/07
EOD blocks:       27,   40, 15/03/07
TKD blocks:        0,    0, 00/00/00
EJ change blocks:  12,   23, 13/03/07

Electronic Journal Status
MEDIUM_PRESENT
MEDIUM_INITIALIZED
MEDIUM_VALID
MEDIUM_AUTHORIZED
```

```
Electronic Journal medium information

Authorized medium
UMID  :        SXXXXXXXXXX03201C43AE
Manu. :                        SanDisk
Desc. :     SD512 Rev:8.0 Date:06/06

Current medium
UMID  :        SXXXXXXXXXX03201C43AE
Manu. :                        SanDisk
Desc. :     SD512 Rev:8.0 Date:06/06

Medium Number:                      12
First Session ID:                   11
Last Session ID :                   27

Initialization Info
Date:                 13/03/07 09:03:00
Fiscal Unit:                T2:TP000002
Firmware version:             33-01-98

Capacity Overview
Nominal Capacity:           495488 KBytes
Usable Capacity:            491376 KBytes
Used space:                    128 KBytes
Free space:                 491248 KBytes

D E B U G
Jmp 1 (TKD) is open
Jmp 2 (Update) is open

Current date & time
16/03/2007      08-21-30
```

4.1 Technical Data

- Technology — High-speed thermal print
- Resolution — 8 dots/mm (203 dpi), 640 dots/line
- Printing speed — one color: 220 mm/s,
- Interface options — 2 x RS232C; USB Modem; USB Host; USB device; Ethernet LAN
- Cutter — Material: tempered steel
- Paper transport — Forward; to use paper to full capacity after cutting:
  up to 12mm backwards
  (approx. 3.5 lines at 7.52 lpi)
- Control functions — Print head temperature control with adjustment of print speed
  Paper Near End Control and Paper End Control
  Paper Cutter error message
  Printer Cover Open/Closed
  Self-test with printout
- Housing color — black
- Power supply — 24 V DC
  Automatic and manual capacity control: 48 – 90 Watt

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- Dimensions 173 x 145 x 195mm (H x W x D)
- Weight 2,84 Kg without paper roll
  3,15 Kg with paper roll
- Features Simple Paper roll exchange:
  Paper near end message: adjustable by user
- Statistical data: Total number of dots
  Total line feeds
  Total number of cuts
  Max. Head temperature
  Paper jam counter
  Cutter error counter
  Thermistor error counter
  High voltage/low voltage error counter
  Number of firmware updates
  Power on time in hours
  Power on counter

- Environmental
- Conditions 5° - 40° Celsius;
- Humidity 5 % - 95% (not condensed)
- Noise <= 55dB(A) operation
- Graphic feature TH230+ is fully graphic-compliant

4.2 Paper Specification

- Paper width 80mm
- Paper weight 55g/m² ± 5 g/m²
- Paper thickness 0.055mm – 0.1mm
- Paper roll
- outer diameter 90mm max.
- Paper length ~100m
- Core size Core diameter: 10mm +2mm
  Wall thickness of the core: max. 2mm ± o.3mm

- Print width 72mm = 576 dot

4.2.1 The main components of the Fiscal Printer are:

- Printer
- Fiscal Controller
- Fiscal Memory

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- Electronic Journal (EJ)
- Data Communication Module – Data Transmission Terminal to ANAF

The last two components (EJ and CoMo) are introduced in the Electronic Journal fiscal law and will be subject to a specific and complex set of tests.

It is recommended for the fiscal POS application to keep the history of the sales and tickets done by the printer, in order to be used by the operator to search and find specific tickets. This journal should be available on the back store server.

4.2.2 Printer sensors:

a. **Paper Near End** – when it is triggered, the PAPER LED of the printer will turn yellow. It detects when the paper is almost ended. The receipt can be closed (as long as MISSING PAPER sensor is not ON). After current receipt is closed, another one **can't** be opened until new paper is put into the printer.
When PAPER NEAR END is triggered during fiscal ticket, the fiscal POS application has two possibilities:
    - ***Cancel the ticket which is in progress, show a message on operator display to change the paper role and after a new paper role is put into the printer, the ticket will be reprinted from the beginning (this is usually specific for hypermarkets environment where the ticket can be very long)***
    - ***Or, the printing/sales can continue until it will be closed (if MISSING PAPER is not triggered before this). After the ticket is closed, the POS application will show a message on operator display.***

b. **Missing Paper** – when it is triggered, the PAPER LED turns yellow and ERROR LED on the printer will turn RED. ***The fiscal POS application must detect this status and must stop sales/ printing***. A message must be displayed on cashier display. Paper needs to be put into the printer and after this the ticket must be canceled and reprinted from the beginning. The reason of this request is that it is not allowed by the fiscal law to have a fiscal ticket on two or more pieces of paper.

When the ERROR LED is RED, the POS is offline related to the fiscal controller but the POS application receive the status of fiscal controller by ASB (Automatic Answer Back) procedure. The POS application can use also the ESC sequence to read the status of the printer using GS_ENQ (paper near end; paper end; cover open; busy; error, etc.)

When Missing Paper sensor will be triggered, the printer enters in PRINTOUT_INTERRUPTED or CMD_INTERRUPTED status.

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The POS application must check the status of the fiscal controller and if the status is in PRINTOUT_INTERRUPTED issue command SPECIAL_ALL_VOID (the fiscal ticket will be canceled) and if the status is CMD_INTERRUPTED issue the command CMD_RESTART.

c. **Cover Open** – it detects when the printer cover is open. ***The fiscal POS application must detect this status and must stop sales/ printing***. Printing will continue only after the cover will be closed.

d. The printer may be powered from the Beetle or by a separate external power supply.

1. ***If the printer is powered from Beetle/POS unit, only the recovery after power off will be tested (printer goes offline together with the Beetle)***
2. ***If the printer is powered on from the external power supply, that will require three set of tests for power off during each area of the ticket:***
   - ***Only for printer powered off. Fiscal POS application must stop the sales. It's not allowed to continue sale without the printer online;***
   - ***Only Beetle/POS unit will be powered off;***
   - ***Both printer and Beetle/POS unit will be powered off.***

4.3 Printer restrictions:

a. Can't sell an article with price zero – it will return error.
b. Can't sell an article with "article name" blank.
c. Can't sell an article with quantity zero

The fiscal POS application must check before sending to the printer these restrictions and must react with a warning on the operator display, otherwise the printer will send back an error which must be handled.
It is required by the law to check the printer status before sending the command to the printer, and we recommend to show an error message on operator display in case of a printer error.
Also for buffer printing it is necessary to check the printer status before starting the receipt. During scan of items the correct information must be displayed on the Customer Display (connected in the printer).

Printer can be used in "scan and print" or in "buffer print" mode.
With each DAY_BEGIN a new EJ file is open. All the fiscal tickets done during the fiscal day will be stored in EJ file (without graphical information). The reports printed on the paper with data from fiscal memory or EJ will not be stored in EJ file. At every End of Day the printer closes the EJ file.

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5. Fiscal Controller: The Fiscal Controller of the printer is used to:

Perform calculations and verifications during a fiscal transaction;
Internal calculations are done using 3 decimals and the values are rounded to two decimals when they are printed on the receipt or report.

The fiscal POS application must use the same way of calculation in order to avoid differences on TOTAL LINE (of the ticket) and TOTAL Daily Sales at RECEIPT END. It will be two special tests (cross checking) to be performed – they can be found in Chapter Special Procedures.

Keep the records of values for fiscal receipts during fiscal day;

POS application will use a special procedure for validation of transaction in case the fiscal ticket was closed – it can be found in Chapter Special Procedures.

Store the Daily VAT values in Fiscal Memory

The EJ fiscal printer has 6 editable VAT rates:

- ***4 mandatory VAT rates (VAT A, B, C and D)***
- ***2 additional/optional VAT rates (VAT G and VAT H).***

POS application must synchronize the VAT rates from Back Store Server to POS Application and also to Fiscal Printer.
To setup or change a VAT Rate, POS application must send to fiscal printer the values (including decimals and without coma separator) for all VAT Rates table:

VAT Rate "A" to VAT Rate "D" – changeable (can't be disabled)
VAT "E" and VAT "F" – always zero (0000)
VAT "G" and VAT "H" – by default are disabled (value = 9999) or other values to enable them if they are used.

Store the end of day (Z Report) values in Fiscal Memory

Store receipts in Electronic Journal

Reprints tickets from Electronic Journal

POS application must have the capability to:
Search and reprint of tickets from Electronic Journal by Date Time and by Session / Z Report number
Request to the printer different EJ Reports and save the information on PC/Beetle

Manages the Data Communication Module to ANAF;

The protocol of transfer data from fiscal printer to ANAF server is done via XML files.
The XML files are created in Communication Module from the data received from fiscal controller.
The data represents information related to the tickets, Z Reports, events, etc.

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The fiscal memory works only in adding mode. It's not possible to decrease/subtract values from the fiscal memory. From this reason, it's impossible to void a fiscal transaction if the fiscal receipt was printed. ***In some POS applications exists a function for CANCEL Ticket after the ticket was closed. In Romanian fiscal environment, this function is FORBIDDEN and it can't be available on the cashier keyboard or touchscreen.***

It is also **forbidden to void an article** that was not previously sold on the fiscal ticket.
Also, it is **forbidden to use the function return and refund article on the fiscal receipt**. For this functions is a special procedure described below in the document.

Handle situations of Fiscal Memory Near Full and Full and EJ Near Full and Full:
Fiscal Memory is Full
When the fiscal memory will be full, the fiscal POS application must display a message on the operator display related to this situation;
It is forbidden to sale if the fiscal memory is full. Message example: "Fiscal Memory is full. Call service support!"
In case of Fiscal Memory Full, the reporting from MF and EJ is mandatory to be available.
Electronic Journal is Full
When the EJ will be Near Full and Full, the fiscal POS application must display a message on the operator display related to this situation;

Recommended: At each fiscal day begin, the POS application must read the fiscal controller status to check if the status bit for Memory near full; EJ near full; Memory full; EJ full was raised.

6. Customer display:

Wincor Nixdorf Customer Display 2x20 characters must be used. Fiscal kit of TH230+MF printer includes a BA64 customer display.
The display must be connected directly in the Fiscal Printer, on a serial port – legal requirement.

When an item is scanned, on the Customer Display must be shown sales information
The POS application must handle the display;

Total line (balance due) must be printed on Customer Display,
The POS application must handle the display;

Each Payment should be displayed on Customer Display and the Balance Due will be refreshed.
The POS application must handle the display;

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It is mandatory for the *Change* value to be displayed on Customer Display until the printing of the ticket ends (close a cash drawer).
The POS application must handle the display;

If using sales printing in Buffer Mode, the items will be displayed on Customer Display only when they are scanned. It's not necessary to display them again when the ticket will be printed, otherwise it will slow down the printer.
The POS application must handle the display;

It is **FORBIDDEN** to sell if the Customer Display is disconnected from the fiscal controller. The Customer Display must be connected to fiscal printer serial COM. **It is not allowed to use a customer display connected to Beetle**.
The POS application must check that the customer display is connected:

- When a new receipt is opened
- Before scanning each item

In case the customer display is not connected, the POS application must return an error on the screen "Customer Display Is Not Connected".
The POS application must handle this error and put a warning message on the operator display

7. Main changes in the current EJ fiscal law:

7.1 Article Sell instruction contains new fields for Measurement Unit and Group of Articles.
Impact in fiscal POS application. The POS application must maintain the item database the Measurement Unit for each item. The Group of Articles is a mandatory parameter that will always be set to "0".

7.2 The quantity of Article Sell can be specified by fraction (e.g. 3/20 pills) – optional;
Impact in fiscal POS application

7.3 Fiscal printer allows the usage of fiscal tickets with zero value.
In case of selling an item with 100% discount, than a fiscal receipt with TOTAL 0 must be issued.
In case of selling an item and then voiding the item, a non-fiscal / voided receipt must be issued.
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7.4 Fiscal ticket with CIF

- A special procedure must be taken into consideration by POS application in order to register the customer CIF (TAX ID).

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- Scenario:

  - During fiscal ticket, the customer may inform the cashier that he wants to add to the fiscal ticket the CIF information (the TAX ID of the company that he represents). The CIF(TAX ID) contains:
    - Optional fiscal attribute – it can be "RO" or "" (blank)
    - TAX ID: between 2 and 13 digits
  - POS application must have the possibility for the cashier to enter from keyboard or touch screen the CIF information (fiscal attribute and TAX ID). POS application can show a check box on the screen for fiscal attribute "RO".
  - POS application must keep this information until the ticket is finished
  - When the ticket is closed, the command for closing the receipt (RECEIPT_END) has a parameter for CIF. CIF information (fiscal attribute together with TAX ID) will be send to fiscal printer together with the command for receipt end.
  - Fiscal printer will keep record for tickets with CIF, sales done on tickets with CIF, VAT values for tickets with CIF to be registered in the fiscal memory.

- For each ticket with CIF, the CIF number (including RO or without RO), brutto sales value, VAT value per each vat rate must be stored and generate a report on server side. ***(More info can be provided by accounting department of the retailer)***

7.5 Media Exchange tickets

- In case the cashier makes a mistake when registers the payments (cash instead of card, or else), it is *optional* that the POS application to provide the possibility to exchange media payment.
- For this function, a function named "Media Exchange" can be available on the keyboard or on the operator screen.
- This operation can be done by POS application only using receipt type = 6 (payment correction) and:

  - PAYMENT_IN function – to introduce the correct media payment
  - PAYMENT_OUT function – to extract the wrong media payment

7.6 ARTICLE_TAX command

- Special command used to collect other taxes in VAT Rate F
- E.g.: Culture Tax, Green Tax, etc..

7.7 Z Report contain more information:

- Management of media payment

  - All media payments will be stored in fiscal memory
  - Dedicated commands for setup media payments
  - Dedicated commands for use of media payment

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7.8 EJ information

7.9 Message Digest SHA 2 information on last line of the Z report

7.10 Additional counters

7.11 Additional GET commands

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7.12 Reports from EJ

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7.13 Management of special files created from information downloaded from:

- Fiscal Memory – extension .MF
- Electronic Journal – extension .EJ

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7.14 Management of XML files generated by the printer in OFFLINE mode, which must be sent to Fiscal Authority.

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7.15 The new fiscal printer has the capability to print a lot of different barcodes: (e.g.: EAN, UPC, Code 39, Code 128, Interleaved 2of5, Codabar, PDF 417, GS1 DataBar)

➢ Barcodes can be printed in sales area, after payment area and before fiscal trailer

Impact in fiscal POS application

7.16 The new fiscal printer has the capability to print in non-fiscal tickets using printing in standard or compressed mode (56 characters / line) and also different printing characteristics as: double high, double width, bold, italic, etc.

Impact in fiscal POS application

7 RECEIPT TYPES:

The receipt type must be specified by POS application every time a new receipt is open

8.1 receipt type = 1: Retail Sales Receipt – Fiscal Receipts

➢ this type of receipt is used for fiscal tickets

NOTE: if the command SPECIAL_ALL_VOID is used during the fiscal ticket, the ticket type will change automatically to receipt type =2 (Generic Receipt – Non-Fiscal) and the fiscal receipt will be canceled by fiscal printer.

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8.2 receipt type = 2: Generic Receipt – Non Fiscal Receipts

- this type of receipt is used for non-fiscal tickets
- the information inside of Non Fiscal receipt will be print using FREEPRINT NORMAL

8.3 receipt type = 3: Special Cash In Receipt – Receipts for Initial Float, Extra Float, Paid-In

- this type of receipt is used like non-fiscal tickets for Special Cash In
- **Initial Float, Extra Float, Paid-In** are commands for cash management which are not related to the sales.

8.4 receipt type = 4: Special Cash Out Receipt – Receipts for Pick-Up, Paid-Out

- this type of receipt is used like non-fiscal tickets for Special Cash Out
- **The printer firmware version supports Pick-up only for cash media payment. Will be in charge of POS software application to do the management of other media payment (except Cash) at the Cashier Declaration.**

8.5 receipt type = 5: Retail Sales Receipt with CIF – Fiscal Receipt with CIF

- this type of receipt is used for fiscal tickets with CIF
- CIF means customer Tax ID. At request of the customer, the cashier receive from the customer the data of the CIF and the receipt will be switch from fiscal normal receipt to fiscal receipt with CIF (simplified invoice)

NOTE: this kind of ticket usually begin as receipt type = 1 (fiscal receipt), but at the end it will receive the CIF information in RECEIPT_END command, and it will be closed as receipt type = 5

8.6 receipt type = 6: Payment Correction – Receipt for media payment

- this type of receipt is used for non-fiscal tickets for “Media Exchange” function

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8 Fiscal ticket – General points:

9.1 Ticket Headers

9.1.1 Graphical Logo Header

- Fiscal printer has the capability to store and print the graphical logo (e.g. company logo);
- It will be in charge of **service tool application** to setup the graphical logo in the fiscalization process and also it can be loaded / changed at request of the retail customer

9.1.2 Fiscal Header

- Printing of the fiscal header is the responsibility of the fiscal printer
- It is mandatory for each receipt/ticket and reports
- It will be in charge of Authorized Service Company to setup the fiscal header using the service tool.
- Fiscal Header has 3 to 5 lines (44 characters per line)
- Lines 1, 2 and 5 are mandatory. Lines 3 and 4 are optional – if they are blank, they will be nor printed.
- Line 1 – Company Name
- Line 2 – Address of the store
- Line 3 and line 4 – extensions for address, if needed.
- Line 5 – CIF of the company
  a. First 25 characters from Line 5 can be modified. Example: "Cod Fiscal C.I.F.:"
  b. The next 2 characters will be Fiscal Attribute: "RO" or " "
  c. The next 13 characters will be for Company CIF
  d. The fiscal attribute and CIF will be automatically extracted by fiscal printer from SNO block in 9.1.2.5 fiscal memory and printed on the last Header Line.
- The fiscal header can be changed 16 times. After this, Fiscal Memory and EJ SD Card must be replaced with new ones in order to allow other changes.
  If CIF of the company will be changed, Fiscal Memory and EJ SD Card must be replaced.

9.1.3 Additional header lines

- **Fiscal POS application** can use "Free Print Restricted" command in order to print additional header lines. E.g. like "store working hours", hello messages, etc.
- In Free Print Restricted can't use numbers is positions from 33 to 44;

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9.2 Sales Area – the area of the receipt where the sold items are printed

- It starts with printing of the currency (LEI) on the right side of the ticket, before first sales line. This is in charge of the fiscal printer to print.
- For each item sold, the POS application sent to the fiscal printer the **Article Name**, **Unit Price**, **Measurement Unit**, **Quantity** of the article, **VAT rate ID** and ***Group of Articles***.
- The **Group of Articles** must be sent as parameter for the Command ARTICLE_SELL, and it has to be set to value "0". It will not be necessary to have the Group of Articles maintained in the item database.
- For each item sales, the command ARTICLE SELL can accept and print on the ticket a **POST LINE** (e.g. comments, article name in other language, article EAN digits code, etc.). This is in charge of fiscal POS application.
- The total value of the article will be **calculated by the fiscal controller working with 3 decimals** and it will rounded to two decimals when it will be printed on the fiscal receipt.
- In order to check the Total Ticket Value, in the same time the POS application makes also its calculations for the value of article.
- Fiscal printer works with the **quantity on 3 decimals** (exception to this rule are the fuel items which can use special command and 5 decimals). Quantity can be also specified as a fraction. The POS application must work in the same way.
- At the end of the sales, the fiscal controller will have the total of the values for all articles in the receipt and the POS application will have its own calculated total value. When a TOTAL command is issued by the POS, this contains the total value calculated by the POS, if this value does not match the value calculated by the fiscal controller then the printer will cancel the receipt (it is mandatory that the POS reads the current receipt total from the fiscal controller before sending the TOTAL command).
  One special procedure is described further in the document at special procedure chapter.
- It is **not allowed to do sales on Group/Department**. Only sales per/as item are allowed;
- Promotions are allowed
  - Promo messages
  - Promo on Item
  - Promo on Ticket Total;
  - Group Promotion / Mix and Match Promotions will be handled just as Ticket Total promotions
  - Fiscal POS application must calculate and send to the fiscal printer the value of the discount (even for Percentage Discount);
- Loyalty cards are allowed to be used in order to get promotions on item or on ticket total;
- Surcharges are allowed (e.g.: additional cost for night sales, for cold drinks, etc.)
- Aritmetical rounding rules apply:
  - <0,5 -> 0
  - ≥0,5 -> 1

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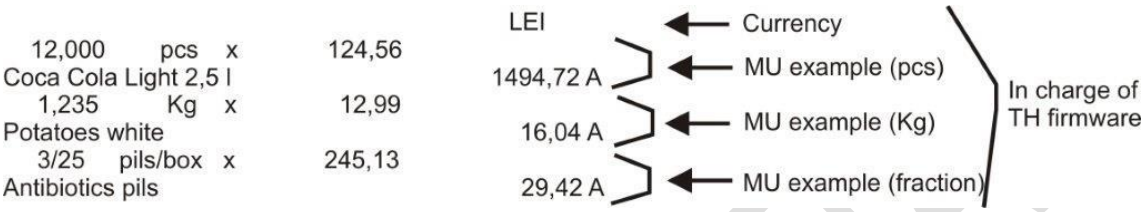
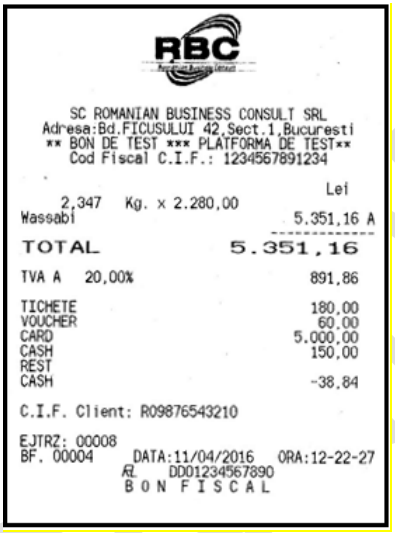
- Article sell commands:

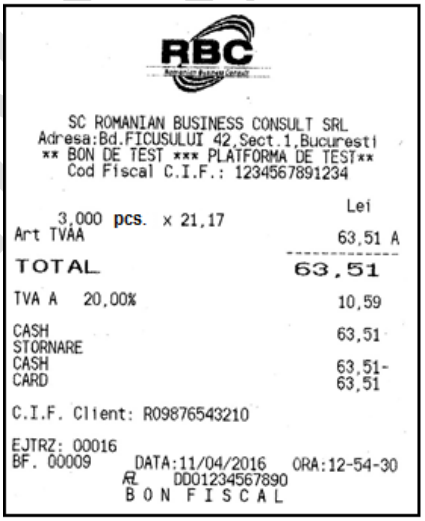
New Parameter **Measurement Unit**

- Represent the Measurement Unit which must be printed on the ticket
- E.g.1: 1,434 Kg x 4,34
- E.g.2: 2,000 pcs. x 3.49
- Quantity unit: max 5 bytes





E.g.: Measurement Unit –
"Kg" printed on the fiscal ticket



E.g.: Measurement Unit –
"pcs." printed on the fiscal ticket

- **New Parameter Fraction from Quantity Box** - Used for selling fractions from Box

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E.g.:
20/100 pills x 21,00
Pills 4,20 A

SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

Lei
5/ 10 Pills x 3,80
Paracetamol Ozone 1,90 A
4/ 20 Pills x 4,90
Acid acetilsalicilic Ozone 0,98 B
TOTAL 2,88
TVA A 20,00% 0,32
TVA B 9,00% 0,08
CARD 2,88
EJTRZ: 00003
BF. 00002 DATA:25/04/2016 ORA:12-20-34
AL DD01234567890
B O N F I S C A L

E.g.: Measurement Unit – "Pills" (pills from a box) printed on the fiscal ticket

9.2.1 Fiscal ticket with TOTAL zero

MF-TH230+ RO allows fiscal ticket with balance due = 0.00

SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

Lei
1,000 buc x 3,00
Art TVAA 3,00 A
DISCOUNT 3,00-A
TOTAL 0,00
EJTRZ: 00005
BF. 00004 DATA:26/04/2016 ORA:10-18-23
AL DD01234567890
B O N F I S C A L

E.g.: Fiscal Ticket with value 0.00

9.2.2 ARTICLE_RETURN

- According to the Romanian Fiscal Law, fiscal printer cannot register negative amounts into fiscal memory. From this reason a special procedure will be used for Return of Articles.
- The Romanian Fiscal Law explain on art. 37:

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In case of RETURN of goods, the User made a Paper File with the followings:

a. **Note of reception** (cod 14-3-1A), on the base of fiscal receipt issued in the past for goods (when they were sold).
b. **Payment Order** to central cashier (cod 14-4) used to prove that the customer received his money back.
c. ***Mandatory*:** Invoice or Invoice in RED (Negative Value) in order for the retailer to collect the VAT for returned articles.

- When an article is returned, a fiscal document has to be generated: a negative invoice (Invoice in Red).
- It isn't possible to print on the fiscal printer a fiscal receipt with a negative value => an invoice with negative quantities, value, total and VAT has to be generated, even if the client doesn't want the invoice.
- All these return invoices will be collected in the accounting department and are the base of returned VAT and 394 monthly tax declaration.
- 2 documents (Payment Order and Note of Reception) are generated on the laser printer or fiscal printer as non-fiscal tickets. The customer has to sign the 2 copies in order to prove that the product has been refunded and the money has been given back to the customer. The 2 documents are kept by the cashier.
- The returned articles can be registered in the system for stock management in the back office software. Therefore, in this case the returned articles must be entered or scanned by the cashier and recorded in the export file for the back office.
- A non-fiscal receipt has to be issued for returned articles. This is a non-fiscal receipt used as an internal document of the store (not given to the client).
- All the documents layout and a possible solution to implement at the POS level the return procedure (using non-fiscal tickets printed on fiscal printer) are available in RBC deliverables.

9.2.3 ARTICLE_FUEL_SELL command

- The fiscal printer has the capability to sell Article Fuel specific for Gas Stations
- For this type of business there are some specific features:
  a. The quantity is working with 5 decimals in order to match the calculations of the fuel pump.
  b. Linked to the fuel item can be a special tax (Road Tax), registered in VAT "F" (Other Taxes).

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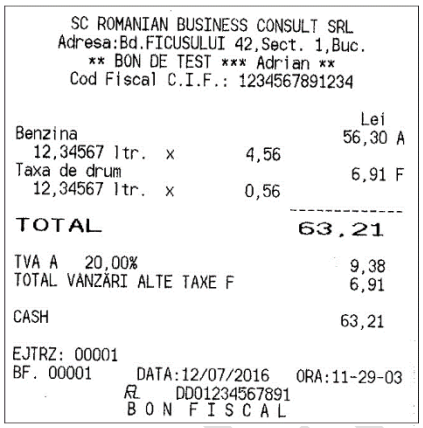
c. The Road Tax may have zero value, or other value and it is specified per liter of fuel. If Road Tax has value zero, it will not be printed on the fiscal ticket.

d. The value of road tax must also be included in Single Price even if it is specified separately in Road Tax (eg.: if fuel price is 4 Lei/liter and road tax is 0.5 Lei/liter, then parameter Single Price will be "450" and parameter Road Tax will be "50").

Note:

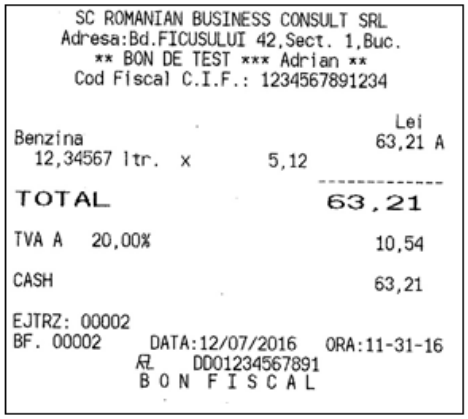
Currently, Road Tax is zero in Romania

If the fiscal ticket contain FUEL_SELL_ARTICLE, then it is forbidden to use SUBTOTAL_DISCOUNT (total ticket discount) because we cannot give discount to tax. In this situation, it is recommended to use after SUBTOTAL the command ARTICLE_DISCOUNT.


SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect. 1,Buc.
** BON DE TEST *** Adrian **
Cod Fiscal C.I.F.: 1234567891234

Lei
Benzina 56,30 A
12,34567 ltr. x 4,56
Taxa de drum 6,91 F
12,34567 ltr. x 0,56
TOTAL 63,21
TVA A 20,00% 9,38
TOTAL VANZARI ALTE TAXE F 6,91
CASH 63,21
EJTRZ: 00001
BF. 00001 DATA:12/07/2016 ORA:11-29-03
AL DD01234567891
B O N F I S C A L

E.g.: Article Fuel Sell with Road Tax


SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect. 1,Buc.
** BON DE TEST *** Adrian **
Cod Fiscal C.I.F.: 1234567891234

Lei
Benzina 63,21 A
12,34567 ltr. x 5,12
TOTAL 63,21
TVA A 20,00% 10,54
CASH 63,21
EJTRZ: 00002
BF. 00002 DATA:12/07/2016 ORA:11-31-16
AL DD01234567891
B O N F I S C A L

E.g.: Article Fuel Sell without Road Tax

9.2.4 ARTICLE_TAX

- Can be used for collecting TAX
- All the amount is registered on VAT F – Other Taxes
- E.g.: culture tax, green tax, visitor's tax, etc.

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Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

	Lei
Article TAX	9,30 F
TOTAL	9,30
TOTAL VANZARI ALTE TAXE F	9,30
CASH	9,30

EJTRZ: 00005
BF. 00005 DATA:26/04/2016 ORA:14-00-19
AL DD01234567890
B O N F I S C A L

E.g.: Article Tax

9.3 Subtotal Area

9.3.1 Subtotal Line – can be used also Subtotal POST LINE

9.3.2 May print Subtotal Discount/Surcharge (exception for ticket containing fuel sell article)

9.3.3 May print Subtotal Messages

9.3.4 Subtotal command differs from TOTAL command as the ticket subtotal value calculated by the POS is NOT included in the command

9.4 Total Area – Balance Due

- Total area contains:
  - TOTAL LINE printed double high, double width
  - Detailed ticket VAT values per VAT rates

➢ Total area information and values are printed by the fiscal printer when receives the command TOTAL LINE from fiscal POS application;

IMPORTANT NOTE: Mandatory Cross Checking of Ticket Total:

- ***In order to be sure that Ticket TOTAL Value is correct, the POS application must read using GET commands, the total value calculated by the printer.***
- ***If the value read from the printer is different from the value calculated by the POS application, the ticket must be canceled.***
- ***An error message must be shown on operator display - message "Fiscal Total Amount ≠ Total POS Amount ".***
- ***No reprint in this situation – a reprint in this case goes to loop!!!***

➢ Total line together with the VAT lines will be in charge of fiscal controller to print. No additional information must be printed by POS software application related to total and VAT.

➢ ***After TOTAL line, the fiscal ticket enters in "Tender Area" (payment area) and cannot go back in the sales area if the customer wants to buy more articles or if the customer has no money to***

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cover the total value if the ticket. From this reason, the management of payment until end of ticket must be in charge of POS application software and the commands for TOTAL and Payment will be send to printer only when the total incoming money is over the total ticket value.

9.5 Tender Area / Media Payment Area

- In the Romanian EJ legislation it is specified that **all the information and values related to media payments must be stored in fiscal memory**, so all management of payment will be in charge of POS application using dedicated payment commands (e.g.: SET_PAYMENT_LIST, PAYMENT_IN, PAYMENT_OUT, etc.).
- Fiscal printer can use up to 20 different media payments:
    - 9 default media payments: CASH, CARD (Credit / Debit / Food Stamps Card), TICHETE (Food Stamps), VOUCHER, BON/TICHET VALORIC (Discount Stamps), CREDIT (Retail Credit), AVANS IN NUMERAR (Cash Back), METODE MODERNE de PLATA (Modern Payment Methods), ALTE METODE MODERNE de PLATA (Other Modern Payment Methods)
    - 11 additional optional media payment: can be defined using command SET_PAYMENT_LIST;

    Note:
    - **Will be mandatory to synchronize the media payment list of the printer with the media payment list of the POS software application.**
    - **The synchronization must be made before DAY OPEN using GET_PAYMENT_LIST command**

- In the tender area, it is mandatory to print the "Change" value (Romanian word for change is "REST"). This value has also to be displayed on the customer display and on the cashier display.

    IMPORTANT NOTE:
    - ***Change Value will be in charge of POS application to be calculated and printed using command PAYMENT_OUT;***
    - ***In order to be sure that Change Values is correct, the POS application must read using GET command the Change Value calculated by the printer.***
    - ***If the value read from the printer is different from the value calculated by the POS application, the ticket must be canceled.***
    - ***An error message must be shown on operator display***
    - ***No reprint in this situation – a reprint in this case goes to loop!!!***
    - ***The fiscal controller will close the ticket only if SUM of ALL Payments minus the Change is equal to Receipt Total***

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The Change that is calculated in the Fiscal Printer can be obtained in the POS application by using the following formula:

Fiscal Printer CHANGE = SUM(ALL MEDIA PAYMENTS) – RECEIPT TOTAL

The list of all media payments for the current receipt can be obtained using function GET_RECEIPT_PAYMENT.
The total of the current receipt can be obtained by using command GET_RECEIPT_TOTALIZER.

9.6 Final ticket messages

- After the printing of CHANGE Line, the ticket can contain some messages. Ex.: thanks' messages; promotion and marketing messages; contact information, etc.
- Regarding the CHANGE line:
- It is recommended that a blank line to be printed before and after the CHANGE line in order to be more visible change information for customer
- These lines are printed in Free Print Normal mode and it will be in charge of POS application;
- In this area, a text can be printed (44 characters / line). Also, barcodes are allowed to be printed in this area;
- The following information is mandatory to be printed just before fiscal trailer:
    - ***POS number***
    - ***Cashier ID or Name***
    - ***Transaction ID from POS application – it's recommended***
    - ***Store number – it's recommended***
- It is recommended to have one blank line before fiscal trailer.

9.7 Fiscal Ticket footer

- The fiscal ticket footer is in charge to be printed by the fiscal printer, after it receives the command RECEPIT_END
- It contains the following information:
    - EJTRZ – transaction ID per day from EJ journal
    - Unique ticket id code ID UNIC - fiscal serial number + ticket date and time + number of fiscal day + number of fiscal ticket
    - Number of fiscal ticket (from the printer side) – this number is reset to one after each Z report. This number might be different from the transaction ID from POS application
    - Data and Time
    - A line with logotype "*RL*" and fiscal serial number (SNO)
    - A line with text "B O N F I S C A L"

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- After the fiscal footer is printed, the printer will cut the paper/ticket

  - In special situations printer allows the use of a command that skips the cut.
  - RECEIPT_END_MFLINE command:
    a. Command to close the ticket without doing the paper cut
    b. Ticket can continue with information (including graphics or barcodes) printed with FREEPRINT_NORMAL Command
    c. To continue, it can either use the SPECIAL_CUT command to cut the paper, or RECEIPT_BEGIN to start a new receipt. If a new receipt is started without a command for a special cut, the paper will not be cut.

  - **Note:**
    a. If you want to print a barcode at the end of fiscal ticket, you must print the graphical barcode before send the RECEIPT_END command

- Validation of the fiscal transaction: At the end of receipt, the POS application sent to the fiscal printer the command RECEIPT_END.
  - At the receipt end process, the fiscal printer will do the following actions:
    a. print the last five lines of the fiscal footer/trailer of the ticket
    b. increase the fiscal number of the receipt (BF)
    c. increase the DAY_TOTALIZER register with the value of the fiscal receipt
    d. get down the flag "open receipt"
    e. write the ticket into Electronic Journal
    f. cut the paper (only for RECEIPT_END Command, for RECEIPT_END_MFLINE Command the paper is not cut)

IMPORTANT NOTE: Validation of fiscal ticket in fiscal printer and POS application

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- ***At the end of the ticket, a cross checking is mandatory to be done by POS application.***
- ***For every fiscal ticket, the POS application must be sure if the fiscal ticket was registered in fiscal printer in order to validate the transaction in its database, otherwise it will lead to daily values differences between fiscal Z Report and POS application.***
- ***The solution is to check the value of DAY_TOTALIZER, if it's incremented after the end of the ticket. If DAY_TOTALIZER was incremented, it means that the fiscal ticket was stored in fiscal printer and the transaction must be valid.***
- ***If DAY_TOTALIZER is not incremented, it means that the fiscal ticket was not ended in good conditions and POS application must read the fiscal printer status and void the ticket. Please be aware that the fiscal tickets with zero value will not increment the DAY_TOTALIZER.***

10. Media Payments:

- Fiscal Printer Wincor Nixdorf MF-TH230+ RO for Romania, works with following media payments:
  - Default: 9 media payments (can't be modified or deleted)
    - Cash (ro: CASH) – ID 0
    - Credit / Debit / Food Stamps Card (ro: CARD) – ID 1
    - Food Stamps (ro: TICHETE) – ID 2
    - Voucher (ro: VOUCHER) – ID 3
    - Discount Stamps (ro: BON/TICHET VALORIC) – ID 4
    - Retail CREDIT (ro: CREDIT) - ID 5
    - Cash Back (ro: AVANS IN NUMERAR) - ID 6
    - Modern Payment Methods (ro: METODE MODERNE de PLATA) - ID 7
    - Other Modern Payment Methods (ro: ALTE METODE MODERNE de PLATA) - ID 8
  - Additional 11 Optional Media Payments: ID 9 to ID 19

- Optional Media Payments are programmable.
  - Can be defined – cmd: **SET_PAYMENT_LIST**
  - Can be updated – cmd: **SET_PAYMENT_LIST**
  - Can be erased – cmd: **SET_PAYMENT_LIST_VOID**
  - Changes to Optional Media Payment are allowed only when the Fiscal Day is closed

IMPORTANT NOTE:

a. *Before Day_Begin the POS application must check the Payment List table inside of fiscal printer to be the same as the table defined in POS application. This can be done using command GET_PAYEMT_LIST for every optional media payment and if*

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necessary, use SET_PAYMENT_LIST or SET_PAYMENT_LIST_VOID commands to synchronize the tables.

b. *Because each media payment has its own ID inside the fiscal printer and for the default media payments it can't be changed and, on the other hand, inside the POS application other media payments IDs are used / required, it might be needed to implement a post processing table for media payments, in order to make the correct links between media payments from POS application to media payments from fiscal printer.*

- Optional Media Payments have the following parameters:
  - Payment Type – payment ID (9 to 19)
  - Payment Name – max 27 bytes
  - Payment Category
    - **2 – Non Cash - Default Value**
    - 1 – Foreign Cash
    - 0 – Local Cash

- For every definition and update of an Optional Media Payment, an Info Ticket is printed
- For deletion of an optional Media Payment, NO info ticket will be printed.

- During sales / fiscal tickets, it's the cashier's responsibility to register the media payments according to what was received from the customer.
- Cannot be given back to the customers the cash change on official food stamp or gift vouchers.

- Food stamps (food vouchers) can be used by customer only to buy food products.
  In order to check the payment, it is recommended that POS application keep the count for each food item sold (if item data base contain one field for food or nonfood articles) that can be paid with food stamp. The cashier can first scan all the food items and use subtotal command in order to have to total value of food articles.

  In Romania there are also Food Stamps in Debit Card form, but this is assigned to CARD media payment.

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E.g.: SET_PAYMENT_LIST Info Ticket

E.g.: Use of Standard Payment (Cash) and Optional Media Payment (ModPlataOptional_20)

10.1 PAYMENT_IN

a. *Media payments must be registered in the fiscal memory and from this reason, for every usage of a media payment, the POS application must inform the fiscal printer with a command PAYMENT_IN the media payment used and its value.*

10.2 PAYMENT_IN_VOID

a. Command used for payment cancelation
b. In order to maintain the correct balance, the POS application must inform the fiscal printer that a media payment was canceled

10.3 PAYMENT_OUT

a. Command used to specify the change returned to customer
b. Fiscal POS application must do the calculations for change, according with balance due and the payments made.
c. It is the POS application responsibility to print the Change (REST) on the fiscal receipt.
d. A line in FREE PRINT must be used with parameter text REST and on the next line, the command PAYMENT_OUT will print the word CASH and the value of the change.
e. Before printing the lines regarding the change, fiscal POS application must obtain the [REST] (English: CHANGE) calculated by fiscal printer. If the values are identical POS application will print a line in FREEPRINT with the REST, and on the next line will use command PAYMENT_OUT to print the value of the change. The following method must be used in order to obtain the CHANGE from the Fiscal Printer:
Fiscal Printer CHANGE = SUM(ALL MEDIA PAYMENTS) – RECEIPT TOTAL

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RBC

SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

		Lei
2,347 Kg. x 2.280,00		
Wassabi		5.351,16 A
TOTAL		5.351,16
TVA A 20,00%		891,86
TICHETE		180,00
VOUCHER		60,00
CARD		5.000,00
CASH		150,00
REST		
CASH		-38,84

C.I.F. Client: R09876543210

EJTRZ: 00008
BF. 00004 DATA:11/04/2016 ORA:12-22-27
AL DD01234567890
B O N F I S C A L

E.g.: Print the change on the fiscal receipt

f. It is recommended that between last media payment in tender area and the line "REST" to print a blank line in order to make it more visible for the customer.

- It is not allowed to round the "TOTAL" value and the "change" value of the fiscal receipt. (e.g. not allowed to round 15.43 to 15.45 on TOTAL Line or for Change)

- Denominations:
  a. Notes: 1, 5, 10, 50, 100, 200, 500
  b. Coins: 1, 5, 10, 50

Coin 1 ban

Coin 5 bani

Coin 10 bani

Coin 50 bani

Note: 1 Leu

Note: 5 Lei

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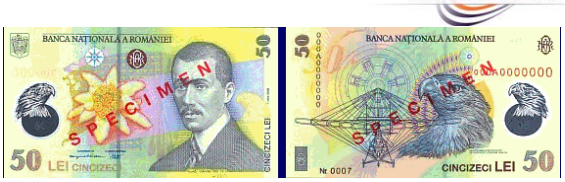
Note: 10 Lei



Note: 50 Lei

Note: 100 Lei

Note: 200 Lei

Note: 500 Lei

- **POS application must handle the management of cash drawer**

10.4 SPECIAL_CASH_IN tickets

- ***Special Cash IN and Special Cash OUT operations must be registered in fiscal memory and from this reason, for every usage of such an operation, the POS application must inform the fiscal printer.***
    - It is ***mandatory*** to do the management of service money using the following functions:
        - INITIAL FLOAT
            a. Money received by cashier in the shift opening, used for change.
        - EXTRA FLOAT
            a. Additional money in the cash drawer during the shift.
        - PAID IN
            a. Money received in cash drawer not related to the sales.
        - PICK UP
            a. Money picked up from the cash drawer, not related to the sales.
        - PAID OUT
            a. Money used from cash drawer to do some payment in store, not related to the sales.
- Every action related to service money must be registered at POS. At the POS side there must be available a function for "**Cashier Report**" which will print a non-fiscal receipt named "Cashier Report" (ro: Raport Casier) which contain a cash drawer balance.
- In the Cashier Report must be printed also the Initial Float, Extra Float, Paid In, Pick Up, Paid Out of the cashier. The reason for this requirement is that the fiscal authority come in front of the cashier and order "stop sale", print a cashier report and check the cash drawer money.

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10.4.1. INITIAL FLOAT (ro: Sold Initial) – CashDrawBalanceType = 0

SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

**** BON NEFISCAL **** BON NEFISCAL ****

INTRODUCERE IN CASA
NUMERAR: SOLD INITIAL
350,00 Lei

EJTRZ: 00001
BNF.00001 DATA:08/04/2016 ORA:12-17-48
DD01234567890
**** BON NEFISCAL **** BON NEFISCAL ****

E.g.: Initial Float ticket

After LOGON, the cashier must make their FLOAT at the POS side (not at the server side). According with this float, **it is mandatory** to print a non-fiscal receipt with the mention of the float like "Sold initial" (in Romanian Language).

10.4.2 EXTRA FLOAT (ro: Alimentare Extra) - CashDrawBalanceType = 1

SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

**** BON NEFISCAL **** BON NEFISCAL ****

INTRODUCERE IN CASA
NUMERAR: ALIMENTARE EXTRA
180,00 Lei

EJTRZ: 00010
BNF.00006 DATA:08/04/2016 ORA:13-59-02
DD01234567890
**** BON NEFISCAL **** BON NEFISCAL ****

E.g.: Extra Float ticket

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10.4.3 PAID IN (ro: ÎNCASĂRI) - *CashDrawBalanceType = 3*

SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

**** BON NEFISCAL **** BON NEFISCAL ****

INTRODUCERE IN CASA
NUMERAR: INCASARI
170,00 Lei

EJTRZ: 00011
BNF.00007 DATA:08/04/2016 ORA:14-00-07
DD01234567890
**** BON NEFISCAL **** BON NEFISCAL ****

E.g.: Paid IN Ticket

10.5 SPECIAL_CASH_OUT tickets

- These tickets are special, and are used to register the "service" money, like PICKUPS and PAID-OUTs
- Can be used only for **receipt type = CASH_IN_OUT (4)**

10.5.1 PICKUPS (ro: REMITERE) – ***CashDrawBalanceType = 2***

SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

**** BON NEFISCAL **** BON NEFISCAL ****

EXTRAGERE DIN CASA
NUMERAR: REMITERE
-5.000,00 Lei

EJTRZ: 00012
BNF.00008 DATA:08/04/2016 ORA:14-02-03
DD01234567890
**** BON NEFISCAL **** BON NEFISCAL ****

E.g.: Pick UP Ticket

10.5.2 PAID-OUT (ro: PLĂŢI) - *CashDrawBalanceType = 4*

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```
SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

****  BON NEFISCAL  ****  BON NEFISCAL  ****

EXTRAGERE DIN CASĂ
NUMERAR: PLĂȚI
-450,00 Lei

EJTRZ: 00009
BNF.00005      DATA:08/04/2016    ORA:13-57-24
DD01234567890
****  BON NEFISCAL  ****  BON NEFISCAL  ****
```

E.g.: Paid Out Ticket

10.6 MEDIA EXCHANGE

a. In case the cashier makes a mistake when registers the payments (cash instead of card, or else), it is ***optional*** that the POS application to provide the possibility to exchange media payment before Cashier Declaration.
b. For this functionality it is required to have a function named "Media Exchange" available on the keyboard or on the operator touch screen.
c. This operation can be done by POS application only using receipt type = 6 (payment correction) and:
    - PAYMENT_IN function – to introduce the correct media payment
    - PAYMENT_OUT function – to extract the wrong media payment

```
SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect. 1,Buc.
** BON DE TEST *** Adrian **
Cod Fiscal C.I.F.: 1234567891234

****  BON NEFISCAL  ****  BON NEFISCAL  ****
                                      Lei
CASH                                -63,21
CARD                                 63,21

EJTRZ: 00003
BNF.00001      DATA:12/07/2016    ORA:11-39-59
DD01234567891
****  BON NEFISCAL  ****  BON NEFISCAL  ****
```

E.g.: Media Exchange ticket

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10. 7 Other info related to payments:

- Maximum allowed for CASH payment:
  - The POS application must have the capability to setup the maximum payment with CASH which is allowed to be received by the cashier. Recommendation is to have it as changeable parameter, in order to be easily changed if the future law will change it.
  - It is ***mandatory*** to have two limits of maximum cash payment:
    1. One for individuals (person): 10.000 lei
    2. One for companies:
       2.1 5.000 lei
       2.2 10.000 lei in Cash & Carry stores
  - If a bigger cash payment is tried on POS, the POS application must prompt a message and not allow the payment.
  - The cashier must check with the customer if it's a company or an individual (person).
  - The payment limit for company (10.000 lei) will be allowed only if the fiscal ticket contains CIF (fiscal TAX ID of the company);
  - The limits are for company (for the entire day and chain of stores from the same retailer).

Fiscal POS application must handle these values and put a warning on operator display.

10.8. Card payment

- The media payment "Manual Card" must be implemented even if we have also the online connection between POS application and EFT terminal. This used when the online connection is broken, or the EFT terminal associated to this POS doesn't work and the cashier use other EFT terminal from other POS check-out.
- Online Card Media Payment – the EFT is connected to POS application
- Two situations can be present:
  a. Direct connection – EFT connected to POS using a specific protocol
  b. Direct connection – using OPI EFT interface (for unattended EFT without paper roll for card transaction ticket and settlement report). In this situation, the POS application must manage:
     - Print payment details related to the EFT transaction (can be included in the fiscal ticket as a free print message, or can be printed as a separate non-fiscal receipt)
     - Display on the cashier display the EFT messages

It will be in charge of the retailer to choose which kind of solution will use, which bank can cover the VISA/MasterCard certification and which EFT terminal will be used.
After this, the specific documentation protocol can be provided by the bank in order to be implemented.

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10.9. Gift Cards:

- In Romania it is allowed to:
  a. Sell Gift Cards as a regular item (including VAT)
  b. Use them:
     - In the sales area of the fiscal ticket as discount voucher – Single Purpose Vouchers
     - In the tender area of the fiscal ticket as media payment – Multi Purpose Vouchers
- The handling of the VAT can be done in the following way. If the store sells items:
  a. From the multiple VAT Rates:
     - The Gift Card will be sold with VAT zero.
     - And it will be used in tender area as media payment (multipurpose mode).
  b. From a single VAT Rate:
     - The Gift Card will be sold with VAT in stores where all items are associated with a single VAT rate (currently 19%).
     - And it will be used in sales area as discount voucher with the same VAT rate.

10.10 E-Gift Cards:

- This is the Electronic Gift Card media payment. A country server or store server must exist, which manages the virtual money balance assigned to the Card ID.
- The Card ID can have a barcode and can be scanned by the cashier when Gift Card is used.
- E-Gift Card can be used in two situations:
  a. Load money (non-fiscal ticket – Paid In) into the money account attached to Card ID. In this case, the cashier receives money from the customer.
  b. Like media payment in tender area. In this situation, the cashier receive money from the money account attached to Card ID

In both situations, POS application must inform the fiscal printer related to management of media payment

- Will be in charge of POS software application to access the country/store server according with protocol of communication (Ex.: INGevoucher protocol)

10.11 Vouchers:

- The voucher is printed on the paper inside the store or outside the store.
- Can be used as media payment

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10.12 Coupons:

- They are created and printed by the POS application on non-fiscal tickets by the fiscal printer
- Can be used like media payment coupons printed inside or outside the store
- If the store uses a 2D Scanner (image scanner), the coupon can also be scanned from phone screen.
- Value can be shown in % and value
- A special coupon management per /customer/country can be implemented

10.13 RVM Coupons:

- POS application can use barcode coupons issued by RVM as media payment
- Some applications take in consideration unique coupons management.

10.14 Cash Back:

- A new law related to Cash Back solution was approved in Romania and it requires the retailer to give back money from EFT terminal (like ATM bank procedure).
- The limit for cash back is maximum 200 lei or total amount of cash in the cash drawer
- This limit must be set as parameter. It might be changed later on.
- A special procedure to manage CASH BACK will be described further in this document.
- When the CASH BACK media payment will be used with PAYMENT_OUT command, in order to declare the amount of cash back given to the customer, the fiscal printer will automatically decrease the Cash in Drawer register with the cash back amount.

Method for implementation – Proposal

1. The items are scanned by the cashier – Ex: scan 1 item of 1 RON
2. The customer mentiones that he want CASHBACK – Ex: cashback amount 200 RON
    a. In the POS application, the cashier selects the function CashBack and introduces the amount (maximum allowed 200 RON)
3. The cashier can continue to scan other items – Ex: another item is scanned with price 2 RON
4. The cashier presses TOTAL for going to the Media Payments
5. The cashier presses the CARD button
    a. Note: there are allowed partial media payments with different media payments than CARD, as long as the CASHBACK amount is higher or equal with the CARD payment amount
6. From the POS application it is sent to the EFT terminal a transaction with 2 amounts:
    a. Amount for the payment of goods (ex: 3 RON)
    b. Amount for CASHBACK (ex: 200 RON)
7. A fiscal ticket is issued where an item named CASHBACK (translated “Avans in numerar”) is printed in the sales area. The payment is done via CARD
8. Another non-fiscal receipt is automatically issued by the POS application, where the following information is printed:

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a. Details regarding the CASHBACK transaction (received from the EFT terminal)
b. Payment in cash done to the customer as a Pay Out

10.15 Mobile Media Payment ("Blink")

- Mobile Media Payment can be used according with specific protocol in connection with the bank transaction server.
- The payment will be downloaded from the account of customer.
- It requires a smartphone running a dedicated application.
- For implementation, a specific protocol documentation can be provided.

10. 17 Other media payments used in Romania:

- Payment by points - Points earned using loyalty cards
- Payment by bank transfer
- Payment with Credit Note

11. Payment in advance

1. In Romania is required to make a fiscal ticket and invoice for payment in advance.
2. It is recommended to create an article named "Avans" with variable price.
3. Because it is not possible to return Advance Payment using the fiscal devices, a special procedure must be applied:
    - The customer orders some articles and pay in advance (partial or total amount).
    - For the value which was paid, the cashier will give to customer a fiscal ticket with article "Avans" and an invoice.
    - When the customer will come to the store to pick up the products ordered, the store will have to do these steps:
        a. Cancel the total value of the payment in advance. For this will be used a procedure for return article "Avans" and issue a negative invoice
        b. After the "Avans" was canceled, the customer and the cashier can scan the articles and pay like a normal sale fiscal receipt. At request of the customer an invoice can be done.
        c. A media payment "Avans" will be used for declaring the amount already paid in advance.

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12. NON-FISCAL Receipts:

1. POS application has the possibility to print also non-fiscal receipts
2. Every non-fiscal receipt contains:
    a. Fiscal Header
    b. Non-fiscal content – a lot of lines printed in Free Print Normal (44 characters / line)
    c. Non-fiscal footer with text "B O N N E F I S C A L" on the last line of the ticket. This line is in charge of printer to be printed.
3. Non-fiscal receipts can include barcodes, graphics and different text formats: double high, double width, bold, italic and compressed.

13. Error handling:

1. During the execution of printing both fiscal or non-fiscal receipts, a variety of printer errors can happen (paper jam, cut error, cover open, etc.).
2. POS application must recover after any error. The POS application must check the fiscal printer and fiscal ticket status.
    a. To do this:
        - it can use for example GET/MFC/TEST commands to receive the status of fiscal controller of the printer (MFC_STATUS)
        - or ESC command to receive the printer status (GS ENQ).
    b. The fiscal printer has implemented the ASB (Automatic Answer Back procedure). According with this function, if some status of fiscal controller changes after a user command, then at the end of command the POS application receives a synchronically the updates fiscal controller status. This can be used to analyze the status bits in order to do the next action. Example: PRINTOUT_INTERRUPT or CMD_INTERRUPTED status.
    c. If the status of fiscal controller is PRINTOUT_INTERRUPTED or CMD_INTERRUPTED (that means the printer was interrupted by some reasons) the POS application can read the printer status (not fiscal controller status) using GS ENQ command in order to get the specific printer error status like: Paper Near End; Paper End; Cover Open; etc.
3. POS application can generate fiscal receipts only for positive sales. No return article receipt is allowed.
4. POS application must check the possible error status in case of sending the command to fiscal printer, and react with a message before sending the command to the fiscal printer.
    a. E.g.: if on the fiscal ticket was sold only one piece of a specific item, and after this cashier tries to void 3 pieces of the same item, the POS application must show an error message and do not send the command to fiscal printer.
5. The fiscal POS application must check before sending to the printer these restrictions and must react with a warning on the operator display, otherwise the printer will send back an error which must be handled.

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6. For buffer printing POS applications it is necessary to check the printer status when scanning the articles.

14. Overflow Management

1. During the fiscal certification procedure overflow limits will be tested
2. The following overflow tests will be made and the **POS application must react with specific messages and do not send command to fiscal printer**:
    a. Quantity x Unit Price <= (less or equal) 9,999,999.99
    b. Ticket total <= 9,999,999.99
    c. Total Sales per day <= 42,949,672.95
    d. Total Discount value <= 42,949,672.95
    e. Total Voided value <= 42,949,672.95
    f. Total Corrections (Return) value <= 42,949,672.95 (not allowed for POS application)
    g. Total Surcharge value <= 42,949,672.95
    h. All daily the other counters <= 42,949,672.95

16. Special Procedures:

1. ***Mandatory:*** When POS unit boots, the POS application must start automatically without possibility to access the OS (Operating System) functions.
2. ***Mandatory Cross Checking of Ticket Total:***
    a. In order to be sure that Ticket TOTAL Value is correct, the POS application must read using GET commands the total value calculated by the printer.
    b. If the value read from the printer is different from the value calculated by the POS application, the ticket must be canceled.
    c. An error message must be shown on operator display - message “Fiscal Total Amount ≠ Total POS Amount“.
    d. No reprint in this situation – a reprint in this case goes to loop!!!

3. ***Mandatory Cross Checking of Change (REST) Value:***
    a. Change Value will be in charge of POS application to be calculated and printed using command PAYMENT_OUT;
    b. In order to be sure that Change Values is correct, the POS application must read using GET command the Change Value calculated by the printer.
    c. If the value read from the printer is different from the value calculated by the POS application, the ticket must be canceled.
    d. An error message must be shown on operator display
    e. No reprint in this situation – a reprint in this case goes to loop!!!

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4. ***Mandatory validation of fiscal ticket in fiscal printer, POS application and transaction back store server.***
    a. At the end of the ticket, a cross checking is mandatory to be done by POS application.
    b. For every fiscal ticket the POS application must be sure if the fiscal ticket was registered in fiscal printer in order to validate the transaction in its database, otherwise it will lead to daily values differences between fiscal Z Report and Total Day value of POS application.
    c. The solution is to check the value of DAY_TOTALIZER, if it's incremented after the end of the ticket. If DAY_TOTALIZER was incremented, it means that the fiscal ticket was stored in fiscal printer and the transaction must be valid.
    d. If DAY_TOTALIZER is not incremented, it means that the fiscal ticket was not ended in good conditions and POS application must read the fiscal printer status.

Note:

Please take in consideration that it can be possible to have fiscal ticket with zero value. In this case Day Totalizer, will be the same.

5. ***POS application journal***
    a. After each fiscal day it is recommended that a backup of the EJ file to be also saved on POS.
    b. An electronic journal must be created by fiscal POS application on the Back Store Server in order to be used by the operator to search, find and display tickets on server display screen.

6. ***Copy of fiscal ticket:***
    a. Romanian fiscal law forbids to make a copy of the fiscal receipt. Tickets can be reprinted also from the printer EJ but they will be marked as "BON DUPLICAT DIN JURNAL" and not as "BON FISCAL" (the cashier needs to explain this to the costumer), reprints from back store end server backup should be handled in a similar way.

7. ***Cancelation of fiscal ticket registered in fiscal printer***
    1. Romanian fiscal law forbids to cancel a receipt after it was closed as fiscal receipt
    2. If the cashier closes a fiscal receipt with wrong values or made a manipulation error, the Romanian fiscal law explains (this procedure will be in charge of the store):
    3. " ART. 36: For the correction and cancellation of a registered operation after issuing a fiscal receipt the user will create the following documents:
        a. the written claim of the person that made the wrong operation, with the mention of correct or cancel reason as well as the order number of the fiscal receipt needed to be corrected, the issuing hour and minute of it;
        b. The written decision for the wrong operation correction acceptance, issued by the financial manager, controller or other person which is in charge of the asset management unit;

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c. the copy from the difference note or reception and ascertainment note, issued from the last inventory until above-ground, on which was established the en-detail price of the item or items wrong registered into the electronic cash register;
d. The accounting note which refers to the effect operation."

8. *24 hours Z Report*

1. ***Mandatory:*** After 24 hours from the **first fiscal ticket** for each fiscal day. It's not allowed to sale after 24 hours have passed since first fiscal ticket of the day:
   - The 24 hours period is calculated from the first fiscal ticket.
   - After 24 hours have passed since first fiscal receipt, the POS application must prompt on operator display a message regarding the situation "24 hours Z Report is required!" when a new item is scanned.
   - **Management of 24 hours is done by fiscal printer**. After 24 hours from the first fiscal ticket, the printer will return an error message. POS application must display it on the screen as "24 hours Z Report is required!"
2. Z Report must be performed (usually in manual mode) before continuing the sales.

9. *Cash Management including all media payments requests special internal procedures in the retail store:*

1. ***Floating Cashier Procedure***
2. ***Because the media payment must be stored also in the fiscal memory, the Cash Drawer is linked to the fiscal printer not to the cashier. From this reason a special internal procedure must be designed if the store use "floating cashiers".***
3. ***Multiple Cashier Shifts per day – same like above description***

10. *Synchronize Date and Time between POS application and Fiscal Printer (cmd: SET_DATE_TIME)*

1. Date and Time must be synchronized for all the entire chain:
2. **Back Store Server -> fiscal POS application -> fiscal printer**
3. ***NOTE: Taking into consideration that fiscal data must be sent to Romanian Fiscal Authority server, in real time and the time stamp of the files is important, it is recommended to have the date and time of the Back Store Server synchronized with Real Time Clock (RTC) Server.***
4. Setup the date and time in fiscal printer:
5. Each block of fiscal memory contains a date and a time-stamp
6. Date and Time of fiscal printer **can't be setup before the date and time of the last block** saved in fiscal memory; (usually the last EOD Z report)
7. As a protection to accidentally setup a wrong date in advance, **printer allows a new date and time with a maximum of two days in the future**.
8. Date and Time of fiscal printer must be setup only in **closed fiscal day**, that means the POS application must setup the date and time before the fiscal day is open (DAY BEGIN command)

Important Note:

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To fulfill the above request, it is mandatory NOT to open the fiscal day right after Z Report. Fiscal day must be open when the cashier sings on the till. This allows POS application to have a time gap to synchronize the Date and Time.

Example: EOD Z report can be issue at 23:00 and fiscal open day must be next day at 8:00. The SET_DATE_TIME will be done before DAY_BEGIN at 8:00.

9. When the fiscal POS application opens the fiscal day, it must check if it has the same date and time as the fiscal printer
   - If there is a difference less than a defined changeable parameter than synchronize automatically (default recommended 15 minutes).
   - If there is a difference higher than the defined changeable parameter than POS needs to display a message regarding this:

     e.g.: "Time and date difference between POS and Fiscal Printer
     POS time is: DD.MM.YYYY HH:MM
     Fiscal printer time is: DD.MM.YYYY HH:MM
     Do you want to synchronize POS date and time to the Fiscal Printer?"
   - If No -> date and time are not synchronized
   - If Yes -> date and time from POS are synchronized to Fiscal Printer
     a. POS application must check if the fiscal day is closed, and after this must do the synchronization.
     b. If the fiscal day is open, the POS application must prompt a message "Can't not set time in fiscal day open"

10. Note:

- ***Changing date and time is allowed by fiscal printer only when the fiscal day is closed.***
- ***The date and time must be equal or greater than the date and time of the last written block in the fiscal memory.***
- ***In case of a date entry, which is more than two days later to the current valid date, the fiscal device will return and error and the date will not be set.***
  - ***Date and Time synchronization must be done each time the application starts, cashier sign on, fiscal day opens.***
  - ***This procedure to be done using special privileges – to be blocked for cashier, but to be allowed to supervisor.***
  - ***This system of synchronization must be implemented in such a way to avoid blocking of sales.***

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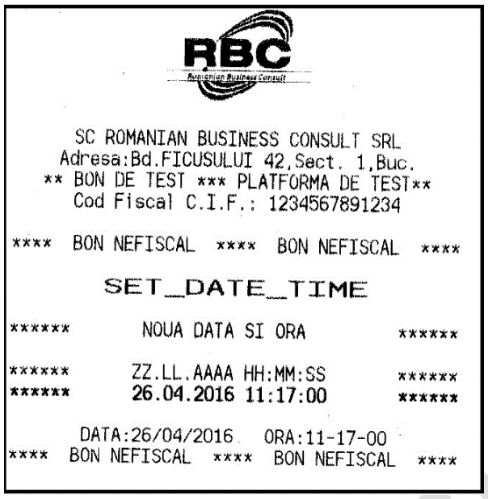
SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect. 1,Buc.
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

**** BON NEFISCAL **** BON NEFISCAL ****

SET_DATE_TIME

****** NOUA DATA SI ORA ******

****** ZZ.LL.AAAA HH:MM:SS ******
****** 26.04.2016 11:17:00 ******

DATA:26/04/2016 ORA:11-17-00
**** BON NEFISCAL **** BON NEFISCAL ****

E.g.: SET_DATE_TIME Info Ticket

11. ***Synchronize VAT Rates between POS application and Fiscal Printer (cmd: SET_VATRATES):***
    1. Setup the VAT Rates in fiscal printer:
    2. VAT Rates must be synchronized for all the entire chain:

       Back Store Server -> fiscal POS application -> fiscal printer
    3. VAT Rates of fiscal printer must be setup only in closed fiscal day, that means the POS application must setup the VAT Rates before the fiscal day is open (DAY BEGIN command)
    4. When the cashier signs on into fiscal POS application, the application must check if it has the same VAT Rates as the fiscal printer.
    5. If there is a difference, POS needs to display a message regarding this:
       e.g.: "There is a VAT Rates difference between POS and Fiscal Printer

       POS VAT Rates are:
       …
       Fiscal Printer VAT Rates are:
       …
       Call the supervisor to synchronize the VAT Rates
    6. Activate supervisor privileges and synchronize the VAT Rates
    7. **Note:**
       a. Changing VAT Rates is allowed by fiscal printer only when the fiscal day is closed.
       b. Fiscal printer allows a maximum of 30 VAT Rate changes. After this, Fiscal Memory and EJ SD Card must be replaced with new ones in order to allow other changes.
       c. VAT Rates enabled by default: VAT A to VAT D (0 to 3)
          - **can be modified**
       d. VAT Rate E (4):
          - used for items which are Excepted from VAT

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- Rate values (%) is always 0.00% **- can't be changed**
- It represents the sales amount of items excepted from VAT

e. VAT Rate F (5):
- Used for Other Taxed (e.g. Road Tax for fuel items)
- Rate value (%) is always 0.00% **- can't be changed**
- It represents the value of other taxes

f. VAT Rates G and H (6 and 7):
- By default are disabled (equal to 99.99%)
- To enable them we need to set up a different value

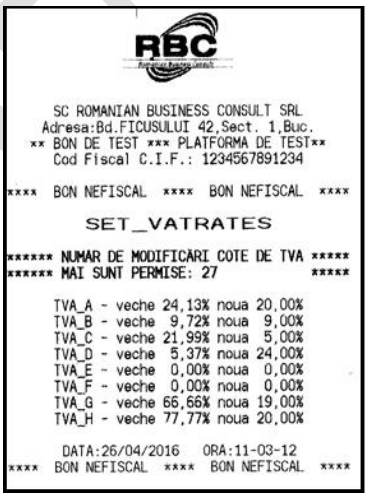
g. It is recommended that, this procedure to be done using special privileges – to be blocked for cashier, but to be allowed to supervisor.

h. To perform a VAT Rate change setup, all 8 VAT Rates must be send from POS application to the fiscal printer, taking into consideration that VAT E and VAT F must be always zero, and if we need VAT G and VAT H to remain disabled must be equal to 9999.

i. If by mistake, VAT Rates E and F will be send to the printer different by zero, the VAT Rates will not be changed.

j. In case the setup of VAT Rates fails, a message must be displayed on operator display regarding this error!
POS application must handle this setup.

k. If 30 changes of VAT is reached, than error MFMEM_BL_LIMIT_OVERRUN is given and no more changes are made. The POS application must stop the sales and prompt the message to "change the fiscal memory" because the VAT rate changes was unsuccessful.


SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect. 1,Buc.
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

**** BON NEFISCAL **** BON NEFISCAL ****

SET_VATRATES

****** NUMAR DE MODIFICARI COTE DE TVA *****
****** MAI SUNT PERMISE: 27 *****

TVA_A - veche 24,13% noua 20,00%
TVA_B - veche 9,72% noua 9,00%
TVA_C - veche 21,99% noua 5,00%
TVA_D - veche 5,37% noua 24,00%
TVA_E - veche 0,00% noua 0,00%
TVA_F - veche 0,00% noua 0,00%
TVA_G - veche 66,66% noua 19,00%
TVA_H - veche 77,77% noua 20,00%

DATA:26/04/2016 ORA:11-03-12
**** BON NEFISCAL **** BON NEFISCAL ****

<u>E.g.: Set VAT Rates Info Ticket</u>
<u>There are still allowed 27 (out of 30) more VAT Rates changes.</u>

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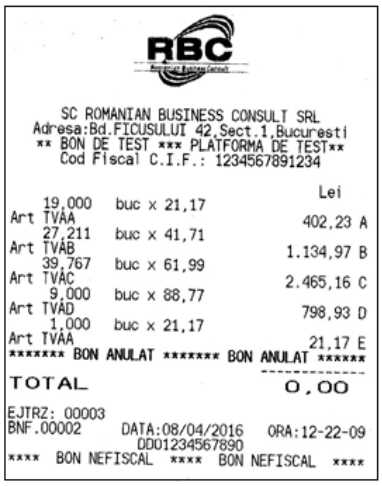
12. SPECIAL_ALL_VOID command

1. This **command is used to cancel a receipt**, fiscal report or different status of the fiscal printer while is still in progress (the command for end receipt was not yet send to the printer).
2. Example/scenario when this command can be used:
   - During sales, we had a power failure. After recovery the printer goes to status "PRINTOUT INTERRUPTED". To recover from this status, POS application needs to use in the first step, the command SPECIAL_ALL_VOID. The receipt will be canceled (it will be closed as non-fiscal ticket) and the status printout interrupted will be disabled. After this, the POS application can reprint the ticket.

NOTE: when the printer status is "PRINTOUT INTERRUPRED", in order to recover, the POS application must use command "SPECIAL_ALL_VOID". If the printer has also "CMD INTERRUPRED" enabled, then the command to recover is "SPECIAL_CMD_RESTART"


SC ROMANIAN BUSINESS CONSULT SRL
Adresa:Bd.FICUSULUI 42,Sect.1,Bucuresti
** BON DE TEST *** PLATFORMA DE TEST**
Cod Fiscal C.I.F.: 1234567891234

Lei
19,000 buc x 21,17
Art TVAA 402,23 A
27,211 buc x 41,71
Art TVAB 1.134,97 B
39,767 buc x 61,99
Art TVAC 2.465,16 C
9,000 buc x 88,77
Art TVAD 798,93 D
1,000 buc x 21,17
Art TVAA 21,17 E
******* BON ANULAT ******* BON ANULAT ******
TOTAL 0,00
EJTRZ: 00003
BNF.00002 DATA:08/04/2016 ORA:12-22-09
DD01234567890
**** BON NEFISCAL **** BON NEFISCAL ****

E.g.: Ticket Voided by Special_All_Void command – ticket is non-fiscal

13. SPECIAL_CMD_RESTART

1. In situation when for some reason (e.g. power failure, paper jam, etc.) the printing is interrupted and the printer status is in "CMD INTERRUPTED", this command must be used in order to resume the command that was previously interrupted.
2. It is mandatory to use it when Z Report was interrupted during printing from different reason. In this situation the printer switch to "CMD INTERRUPTED" and the only solution to recover is to use SPECIAL_CMD_RESTART.
3. From time to time when an error happens while printing the fiscal footer of the ticket, the printer may also go in "CMD INTERRUPTED", and the POS application must use SPECIAL_CMD_RESTART in order to recover.

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NOTE: when the printer status is "PRINTOUT INTERRUPRED", in order to recover, the POS application must use command "SPECIAL_ALL_VOID". If the printer has also "CMD INTERRUPRED" enabled, than the command to recover is "SPECIAL_CMD_RESTART"

14. <u>Read printer status if printer is OFFLINE:</u>

1. In situations when the printer goes offline (e.g.: missing paper, cover open, etc.) and POS application does not receive an answer from the printer, the POS application can use printer commands (not fiscal controller commands) GS ENQ, DLE EOT, GS a) for:
    - Check the status of the printer
    - Please see TH230+ Programmers Guide – page 93

15. <u>Verify ending of previous command in good conditions in order to continue the command</u>

1. **POS application can use command MFC_SYNC (ESC v) which returns the printer status and error message after the previous command was ended.**
2. **Examples:**
    - **After command RECEIPT_END, the printer has a number of activities to perform like: printer trailer, increase the BF number, write in EJ, increase daily totalizer, switch off receipt open flag and paper cut, POS application can use MFC_SYNC in order to receive answer from the printer, at the end of all these activities.**
    - **Take care for printing the fiscal ticket in buffer mode, recommendation is to use MFC_SYNC during the printing of the ticket to be sure that every item was send and printed ok. Otherwise we may have situations like difference between TOTAL calculated by fiscal printer and TOTAL calculated by POS application.**

16. <u>CIF Validation Algorithm</u>

1. CIF represents the company VAT tax number. It contains:
    - „RO" – Fiscal Attribute, used only for company who pay the VAT to the state.
    - „ „ - Blank Fiscal Attribute for company who do not pay VAT to the state.
    - Company VAT tax number – numeric less than 13 digits.
2. **CIF number structure**
    - [|ZZZZZZZZZ|] |C|
    - |_____| |_|
    - : :
    - : --> Check digit
    - :
    - - - > CIF ID (max 9 digits)

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3. The fiscal attribute “RO” is ignored in case it exists.
4. The CIF number is checked to have maximum length of 10 digits and contain only numeric characters.
5. The following test key "753 217 532" is used. The CIF number and the test key are inverted. The inverted test key is “235 712 357”.
6. The first digit of the inverted CIF number is ignored (this is the check digit).
7. Each digit from the inverted CIF is multiplied with the corresponding digit from the inverted test key. (Multiply digit 1 from inverted CIF with digit 1 from inverted test key; multiply digit 2 from inverted CIF with digit 2 from inverted test key, ..)
8. Sum all results from step 5.
9. The sum amount is multiplied by 10 and modulo by 11. The result is the check digit. If the result of the operation modulo 11 is 10, then the check digit is 0.
10. The check digit obtained in previous step is compared to the original check digit from point 4. If they are the same, the CIF number is valid.

A PHP implementation can be found here:

- http://ro.wikipedia.org/wiki/Cod_de_Identificare_Fiscal%C4%83

The printer accepts any CIF shorter than 13 characters to be printed, the POS needs to check the CIF so as not to print illegal CIF's (the law states that the costumer must present a copy of his legal statement that mentions the CIF, the store must make a copy of this statement for later verification, also ANAF will reject any fiscal tickets with illegal CIF).

17. CNP validation algorithm:

The CNP has the following form:
|*S*||*AA*||*LL*||*ZZ*||*JJ*||*ZZZ*||*C*|
|*C*| => Control Digit
|*ZZZ*| => Person serial number
|*JJ*| => County code
|*ZZ*| => Birth day
|*LL*| => Birth month
|*AA*| => Birth year
|*S*| => Gender (M/F) for:
1/2 - Romanian citizens born between 1st January 1900 and 31st December 1999
3/4 - Romanian citizens born between 1st January 1800 and 31st December 1899
5/6 - Romanian citizens born between 1st January 2000 and 31st December 2099

CNP validation algorithm:

- Maximum length of the code is 13 digits.
- For validation is used the following key: 279146358279
- Every digit is multiplied with the corresponding digit of the key

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- The results of all multiplications are added. The sum will be multiplied by 10 and the result will be divided by 11. MOD 11 will be applied - the result represents the control digit. If the obtained result is 10 then the control digit will be 1 (one), else the control digit is the value of the rest.
- In order to have a valid CNP, the control digit has to correspond with the last digit of the CNP code.

18. Data write buffer

- For the POS applications that works with commands directly to the fiscal printer, it is recommended to setup the data write buffer = 4096 bytes, other ways loss of data might happen during printing.

17. Special Menu at POS

17.1. Programming Mode of the POS application

The POS application must have a Menu named "Programming Mode". This menu was requested by ICI institute and it is mandatory for obtaining the fiscal homologation.
All the functions in "Programming Mode" must be **password protected / secured**.
In this Menu the following functions must exist:

a. **Function 1: Edit the VAT rate values on the POS**
   1. The ICI tester needs to select an editable VAT rate (A, B, C or D), and edit the VAT rate value (ex: configure for VAT rate A the value 19.99%, configure for VAT rate B the value 7.24%, etc)
   2. This function must be implemented as a function/button on POS, in the "Programming Mode" Menu
   3. This function must edit the VAT rates in the POS application database. The POS application will handle then the communication with the fiscal printer.

b. **Function 2: Create / edit an existing item on the POS** (function already requested last week)
   1. The ICI tester needs to create a new item or edit an existing item. He must type the code of an item in a GUI interface.
   2. If the item does not exist than the item must be created and the item fields must be edited by the user.
   3. If the item exists than the item fields must be edited.
   4. The following item fields must be edited for an item:
      1. Item Name

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2. Item Price
3. VAT Rate (A, B, C, D or E)
4. Measurement unit – string – max 7 characters long (this field is required only if the POS application supports multiple measurement units)

5. This function must be implemented as a function/button on POS, in the "Programming Mode" Menu
6. This function must edit the items in the POS application database. The POS application will handle then the communication with the fiscal printer.

c. **Function 3: Programming the Date and Time on the POS**
   1. This function must change the date and time on the POS operating system.

d. **Function 4: Fiscalization. This must be a submenu of the "Programming Mode" where the following functions must be present:**
   1. **Programming the Date and Time on the Fiscal Printer**
      1. This function must set the Date and Time in the Fiscal Printer
      2. The direct command that need to be used: **SET_DATE_TIME**
      3. The ProBase command that need to be used: **setDate()**
   2. **Fiscalization of the Fiscal Memory**
      1. This function will write in the Fiscal Printer the Fiscal ID of the Printer, Tax ID of the Retailer (CIF number), and Printer Serial Number
      2. The direct command that need to be used: **INIT_WRITE_SNO**
   3. **Request the pairing between Fiscal Printer and the Tax Terminal**
      1. The direct command that need to be used: **TT_PAIR_REQUEST**
      2. The ProBase command is under development at the ProBase team. An update is expected this week.
   4. **Initialization of the Electronic Journal**
      1. The direct command that need to be used: **EJ_INIT**
      2. The ProBase command is under development at the ProBase team. An update is expected this week.
   5. **Programming the VAT Rates in the Fiscal Printer**
      1. The direct command that need to be used: **SET_VAT_RATES**
      2. The ProBase command that need to be used: **setVatValue()** and **setVatTable()**
   6. **Programming of the Fiscal Header**
      1. The direct command that need to be used: **SET_HEADER**
      2. The ProBase command that need to be used: **setHeaderLine()**

      For setting the header the following inputs for the user are needed:
      1. First line of the header - **mandatory field** - 44 characters allowed - allow the user to type character space (" ") in front of the text, so the user can center the text manually
      2. Second line of the header - **mandatory field** - 44 characters allowed - allow the user to type character space (" ") in front of the text, so the user can center the text manually

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3. Third line of the header - **optional field** - 44 characters allowed - allow the user to type character space (“ ”) in front of the text, so the user can center the text manually
4. Fourth line of the header - **optional field** - 44 characters allowed - allow the user to type character space (“ ”) in front of the text, so the user can center the text manually
5. Fifth line of the header - For this line you have two parameters:
   - The CIF description string - max 25 characters allowed - it will always be set to this value “`Cod Fiscal C.I.F.:`”
     - You can have this string saved in the database / hardcoded / in a configuration file
     - The cashier should not have the possibility to change the preconfigured string “`Cod Fiscal C.I.F.:`”
   - Checkbox for the optional “RO” prefix in front of the CIF number.
     - If the cashier checks it, then the “RO” argument will be send for the parameter CIF_ID. The CIF number that is printed in the Fiscal header will contain the “RO“ Prefix
     - If the cashier does not check it, then a blank “” argument must be sent for the parameter CIF_ID. The CIF number that is printed in the Fiscal header will contain no “RO” Prefix

Note: the numerical part of the CIF number that is printed in the Fiscal Header, will be automatically printed by the printer from the SNO block. You do not send the numerical part of the CIF as an argument for the SET_HEADER command.

e. **Function 5: Import interface for the profiles of the Fiscal Printer**
   1. The POS application must send the command to the printer. The command will “tell” the printer to import the XML file with the new profile from an USB Memory Stick connected in the printer.
   2. The direct command that need to be used:
      TAX_TERMINAL_IMPORT_XML_PROFILE

*Note: **The Programming Mode functions must affect only the Fiscal POS**. It is not relevant what information is sent to the Backoffice server / central systems regarding the items that are created/edited and sold on POS, or regarding the VAT rates that are edited.*
In case of the systems with a central database (in ERP system or Backoffice system), then it is allowed that the central database to overwrite the POS database at EOD. This means that all the local POS changes will have effect only until EOD.

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17.2. Reporting Mode of the POS application

It's ***mandatory*** for POS application to have a special menu **"Fiscal Reports".**

Mandatory: The menu can be accessed only as Supervisor Cashier using security privileges (minimum 4 digits password)

This menu must contain:

1. **Fiscal X-report** (informal character) (overview daily activity)
2. **Fiscal Z-report** (official character) (the fiscal day will be automatically closed when an Z report is requested) (overview daily total) – always printed

   Note: The Fiscal Z report must be issued only with the cashier logged of. This is a protection for resetting the CASH IN DRAWER register of the printer, when the Z report is issued.
3. **Printer Self-Test (cmd: TEST_PRINT_SELFTEST)**
   a. This command prints a self-test including information about the print unit itself, the fiscal section and the EJ section.
   b. The output is either printed on the receipt station or downloaded into POS application and displayed on operator screen.
4. **Fiscal Date to Date Report – mandatory to support usage of parameters:**
   a. Detailed Fiscal Date to Date Report (REPORT_DATE2DATE)

   For this report, the operator must be able to:
   - Print the report on fiscal printer (parameter station = 2)
   - Export the report on an USB stick connected in the printer on the USB Host port (parameter station = 32)

   b. Summary Fiscal Date to Date Report (REPORT_Total_Date2Date)

   For this report, the operator must be able to:
   - Print the report on fiscal printer (parameter station = 2)
   - Export the report on an USB stick connected in the printer on the USB Host port (parameter station = 32)

   c. The following **parameters** are requested by EJ law in order to be printed:

Value	Report Type	Text
Not present	All	TOATE (prints all MF blocks information for period)
0	SNO	SNO (prints only SNO block info for period)
1	EOD	RAPORT Z (prints only Z Report block for period)
2	TKD	TKD RESET (prints only TKD Reset block for period)
3	EJ Change	MODIFICĂRI JURNAL ELECTRONIC (prints only EJ Change)
4	TH	ANTET (prints only Header Changes for period)
5	VAT	TVA (prints only VAT Rates changes for period)
6	CRY	MODIFICĂRI MONEDĂ (prints only currency changes for period)
7	TT	MODIFICĂRI TERMINAL COMUNICAŢIE (prints only communication module changes for period)
8	FW	FIRMWARE CHANGE (prints only firmware changes for period)
9		Reserved

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5. EJ Reports
   a. These reports are read from EJ files from SD Card and printed
   b. **First EJ report that must be available** at POS - **EJ_REPRINT_T2T_BY_DATE_TIME**
      - The application prints a non-fiscal report over a range of EJ data specified by date.

        For this report, the operator must be able to:
        - Print the report on fiscal printer (parameter station = 2)
        - Export the report on an USB stick connected in the printer on the USB Host port (parameter station = 32)
   c. **Second EJ report that must be available** at POS - **EJ_REPRINT_T2T_BY_SESSION**
      - The application prints a non-fiscal report over a range of EJ data specified by session and ticket info. A session is an EOD block number.

        For this report, the operator must be able to:
        - Print the report on fiscal printer (parameter station = 2)
        - Export the report on an USB stick connected in the printer on the USB Host port (parameter station = 32)
   d. Parameters for EJ Reports (hexadecimal ASCII format) for Ticket Types:
      - 0400h: FISCAL_SALES_RECEIPT_WITH_TAX_ID
      - 0008h: DAY_END_REPORT
      - 0001h: FISCAL_SALES_RECEIPT

To print all the fiscal tickets the command must add the filter for FISCAL_SALES_RECEIPT_WITH_TAX_ID and FISCAL_SALES_RECEIPT

Note:
For EJ report you must check the validity of the range: session number and ticket number per session.
If you choose an invalid range, you will get an empty report.

6. Export XML from Fiscal Printer by SESSION
   a. The POS application must be able to send to the Fiscal Printer a command, in order to trigger the XML files export on a USB Memory Stick. The USB Memory Stick is connected in the printer.
      - The XML files will be generated by the printer when the command is sent, and they will be copied by the printer to USB Memory Stick.
   b. The direct command that need to be used: **TAX_TERMINAL_EXPORT_XML_BY_SESSION**

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18. Z Report (end of day) – closing of fiscal day

1. At End of Day, the closing the fiscal day must be done.
2. All the daily counters, totalizers of fiscal printer will be stored in the fiscal memory EOD block.
3. After printing of Z Report, the **Z report is appended to the EJ file**, then the EJ file will be closed, the SHA 2 algorithm is used to calculate the Message Digest.
4. The Message Digest will be stored also in fiscal memory EOD block and will be printed as last line of Z Report.
5. Closing of fiscal day is performed by printing the Z Report on the fiscal printer
6. This report displays the turnover performed during the day and the VAT collected, all media payments used during the fiscal day, all counters and totalizers.
7. After Z Report is printed, **all the daily counters are reset**.
8. In case **PAPER NEAR END sensor is ON, the report will not be printed.** The status of the fiscal printer must be checked by the POS application before sending the Z Report command.
9. In case a Z Report is interrupted during printing, MFC_CMD_INTERRUPTED bit will be set. The POS application must send the command SPECIAL_CMD_RESTART, to resume printing.
10. Z Report command can be issued by:
    a. ***Mandatory:*** the POS application – manually from a special menu on the POS side
    b. The server during the End Of Day process – automatically
    c. ***Mandatory:*** After 24 hours from the **first fiscal ticket** for each fiscal day. It's not allowed to sale after 24 hours have passed since first fiscal ticket of the day:
        - The 24 hours period is calculated from the first fiscal ticket.
        - **Management of 24 hours is done by fiscal printer.** After 24 hours from the first fiscal ticket, the printer will return an error message. POS application must display it on the screen as "24 hours Z Report is required!"
    d. Z Report must be performed (usually in manual mode after) before continuing sales.
    e. Mandatory: at every EOD, an XML file created based on Z Report will be generated by fiscal printer in order to be send to Fiscal Authority server. It will be in charge of the Fiscal Printer (communication module) to generate the XML files. It is not the POS application job to save the XMLs.

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11. Z Report (DAY_END) template:

01234567890123456789012345678901234567890123		44 characters
***** W I N C O R - N I X D O R F *****		Header Line 1
receipt header line 2		Header Line 2
Cod Fiscal C.I.F.: RO1234567890123		Header Line 3
…		
R A P O R T F I S C A L		Z report
Z I L N I C - Z		
	Lei	Currency label
TOTAL VÂNZĂRI BRUTE	3.258,74	Daily Sales Total
Nr. DISCOUNT	12	Number of Discounts Operations
TOTAL DISCOUNT	120,00	Total Values of Discount Operations
Nr. MAJORĂRI	10	Number of Surcharge Operations
TOTAL MAJORĂRI	75,50	Total Value of Surcharge Operations
Nr. CORECŢII	254	Number of Returns Operations
TOTAL CORECŢII	4.102,33	Total Value of Return Operations
Nr. ANULĂRI	31	Number of Void Operations (items and tickets)
TOTAL SUME ANULATE	820,30	Total Value of Void Operations
Nr. BONURI ANULATE	0	Number of Voided Tickets
Nr. ARTICOLE ANULATE	0	Number of Voided Articles
TOTAL ARTICOLE ANULATE	0,00	Total Value of Voided Articles

TVA (A) 21,00%		VAT rate A
VÂNZĂRI BRUTE	1.710,73	Total value of sales on VAT A
VALOARE TVA	359,25	VAT amount of VAT A
TVA (B) 9,00%		VAT rate B
VÂNZĂRI BRUTE	11,88	Total value of sales on VAT B
VALOARE TVA	1,90	VAT amount of VAT B
TVA (C) 7,50%		VAT rate C
VÂNZĂRI BRUTE	16,92	Total value of sales on VAT C
VALOARE TVA	1,18	VAT amount of VAT C
TVA (D) 00,00%		VAT rate D
VÂNZĂRI BRUTE	00,00	Total value of sales on VAT D
VALOARE TVA	0,00	VAT amount of VAT D
TOTAL VÂN. EXCEPTATE TVA	36,02	Total Value of Sales on VAT E = 0% (Excepted from VAT)
TOTAL VÂNZĂRI ALTE TAXE	1.471,20	Total Value of Sales on Other Taxes (including Road Tax)
TVA (G) 02,50%		VAT rate G
VÂNZĂRI BRUTE	2.000,00	Total value of sales on G
VALOARE TVA	50,00	VAT amount of VAT G
TVA (H) 19,00%		VAT rate H
VÂNZĂRI BRUTE	25,00	Total value of sales on H
VALOARE TVA	5,05	VAT amount of VAT H
TOTAL GENERAL		
VÂNZĂRI BRUTE	102,32	Total Sales value
VÂNZĂRI TAXABILE	3.776,52	Total Value of Sales with VAT (A,B,C,D and G, H)
VALOARE TVA	61,05	Total VAT value (A,B,C,D and G,H)

CONTOR ZILE:		
RAPORT Z NUMĂRUL	9	Counter / Number of the current Z Report
MODIFICĂRI COTĂ TVA	1	VAT rate changes
MODIFICĂRI ANTET	1	Number of Fiscal Header Changes
NUMAR RESET TKD	1	Number of TKD resets
MODIFICĂRI MONEDĂ	0	Number of Currency Changes
MODIFICĂRI JURNAL ELECTRONIC	3	No. of Electronic Journal SD Card Changes

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NR. BONURI FISCALE	22	No. of Fiscal Tickets in this session (fiscal day)
NR. BONURI NE-FISCALE	14	No. of Non Fiscal Tickets in this fiscal day
---		
NR. BONURI CU CIF	8	No. of Fiscal Tickets with CIF in this fiscal day
TOTAL VÂNZĂRI CU CIF	580,24	Total Value of Sales on Fiscal Tickets with CIF
VALOARE TVA CIF	61,05	VAT amount of Fiscal Tickets with CIF
ID JURNAL ELECTRONIC	SXXXXXXXXX0335E5F946	Electronic Journal EJ SD Card UMID
DATA INITIALIZARII	21.11.2017 11:16:56	EJ SD Card initialization date and time
NR. MEDIU SD CARD	1	EJ SD Card number for this MF
NR. FISIER EJ	18	EJ SD Card number of files

NR. SOLDURI INIŢIALE	1	No. of Initial Float Operations
SOLD INIŢIAL	100,00	Total Value of Initial Float Operations
NR. ALIMENTARI EXTRA	1	No. of Extra Float Operations
ALIMENTARE EXTRA	500,00	Total Value of Extra Float Operations
NR. REMITERI	1	No. of Pick-up Operations
REMITERE	300,00	Total Value of Pick-Up Operations
NR. ÎNCASĂRI	1	No. of Paid-In Operations
ÎNCASĂRI	14.189,56	Total Value of Paid-In Operations
NR. PLAŢI	1	No of Paid-Out Operations
PLĂŢI	10.113,04	Total Value of Paid-Out Operations

CASH	2.820,70	Total Value of the cash payment - REST (change given to the customer)
CASH ÎN SERTAR	2.010,00	Total Value of the cash payment - REST + CASH_IN[1] - CASH_OUT[2]
CARD	995,82	Total Value of payment with Card
TICHETE DE MASA	5,00	Total Value of payment Food Stamp
VOUCHER	0,00	Total Value of payment Voucher
BON/TICHET VALORIC	0,00	Total Value of payment Discount Stamps
CREDIT	0,00	Total Value of payment Retail CREDIT
AVANS ÎN NUMERAR	0,00	Total Value of payment Cash Back
METODE MODERNE DE PLATA	0,00	Total Value of payment Modern Payment Methods
ALTE METODE MODERNE DE PLATA	0,00	Total Value of payment Other Modern Payment Methods
<Payment Name 10>	5,82	Total Value of Optional Media Payment 5
. . .		
. . .		
<Payment Name 19>	5,00	Total Value of Optional Media Payment 19
EJTRZ: 00037		Electronic Journal Transaction Number in Session
DATA:03/04/2008 ORA:15-42-23		
RL BU47101234567		
R A P O R T F I S C A L		
7E213D14 7631DBB2 4989A09E B5E399FF E850A116 7631DBB2 4989A09E B5E399FF		EJ session checksum (SHA-2 64 characters)

12. DAY_X REPORT:

a. ***Mandatory:*** to have the possibility to manually print X Report of the fiscal printer.

b. Can be available to be downloaded from fiscal printer to PC in electronic format.

c. X Report and all other reports from Fiscal Memory and Electronic Journal can be printed on paper or can be made available to POS application. The filter can be done in print command using Station Parameter:

- Station = 2 – the report will be printed on paper

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- Station = 16 – information will be made available to POS application

d. **If Station = 16, the application can retrieve data from fiscal printer using command MFC_READ_HOST_DATA, in blocks of 512 bytes, until the last block is smaller than 512 bytes.**

e. ***Mandatory: at request of Fiscal Authority server, an XML file created based on X Report will be generated by fiscal printer in order to be send to Fiscal Authority server. This file must be saved also on Beetle/POS by the fiscal POS application to cover the cases when the connection is offline and the file needs to be transmitted later on using other ways.***

19. Communication Module "CoMo"

1. **Components:**
   a. The fiscal printer MF-TH230+ RO contain a communication module named "CoMo".
   b. This module is in charge to collect data from fiscal controller and send data to fiscal authority server.
   c. In this scope, it is equipped with:
      - Communication interfaces
      - Memory SD-Card to temporary store the data before transmitting
      - Smart SC card for cryptographic algorithm and Digital Signature
      - Power processor for management of transmission, XML files creation and offline status
      - CoMo SD Card used for store XML offline data; Smart Card SC with cryptographic module according with the FIPS 140-2 standard-Level 2, able to store an internal digital certificate X.509.V3 with the length of the RSA key at 2048 bits

2. **Data exchange:**

   The Communication Module is responsible for generating the data for ANAF fiscal authority.

3. The fiscal authority servers are online starting with 31st of March 2021.

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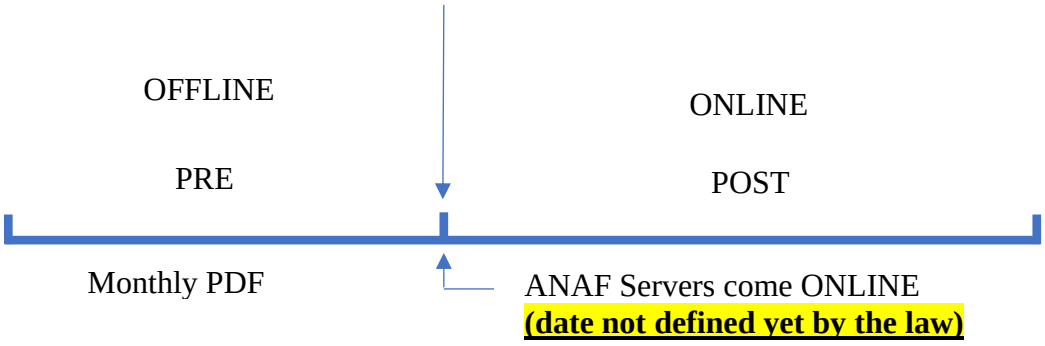

4. Depending on the moment when the Fiscal Printer operates (before or after **ANAF Online Date**) there are two periods of time defined:
    a. **PreOperational** – before **ANAF Online Date**
    b. **PostOperational** – after **ANAF Online Date**
5. The ANAF Online Date will be defined in the future. The current unofficial estimations from the fiscal authorities is about 2 years.
6. The **PreOperational** period:
    a. The Fiscal Printer is NOT connected to the ANAF Server
    b. The Fiscal Printer has **Profile 0 activated** – no connection to ANAF server
        i. The **Profile 0** contains a parameter for the date when the printer will switch to **Profile 1** (printer online to ANAF). At the rollout the printers will have Profile 0 activated.
    c. The Fiscal Printer will generate XML files for all sales and Z reports. It will be in charge of the Fiscal Printer to generate and manage these XML files.
    d. It will be in charge of Retailer Company to send a monthly declaration (until 20th of next month for the previous month), for each printer. The declaration is described by the fiscal authorities in a Smart PDF file where all XML files from the printer must be attached.
    e. The XML files can be extracted from the printer in two ways:
        i. Manually, by the retailer or the service technician, using the XML export function of the POS application. The export is done on an USB stick connected in the printer. A specific procedure will be delivered to the Retailer Company for manually extracting the XML files from the printer.
        ii. Automatically, if the printer will be connected to LAN and will be configured to send the XML files to a server.

        Note: it will NOT be the purpose of the POS application to extract the XML files from the Fiscal Printer. The XML files will be collected by an additional application.
    f. The POS application must NOT check if the printer is offline or not. In PreOperational state, the Fiscal Printer will be always offline from ANAF Server.

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7. The **PostOperational** period
    a. The Fiscal Printer will be connected to the ANAF Server
    b. The Fiscal Authority in Romania must publish all details regarding online communication. All technical details are not known for the moment.
    c. The Fiscal Printer will have **Profile 1 activated** – connection with ANAF server is online
    d. The Fiscal Printer will generate XML files for all sales, Z Reports, events, and the Fiscal Printer will send the files automatically to the ANAF server.
    e. The Profile 1 will have a parameter that will define the number of days until the printer can work in offline mode (offline from ANAF server)
        i. The value for this parameter will be defined in the future by the fiscal authorities
    f. The POS application must check if the printer is offline or not. In case the printer is offline, a message must be displayed on the Operator screen, with the number or days until the printer will be locked. This will be implemented after the fiscal authorities will publish the rules of communication for PostOperational period.

8. <u>Interfaces:</u>

Communication interfaces to the BEETLE or PC
- RS232c
- RS232c with customer display (BA63; BA64)
- USB Modem
- USB Host
- USB device

▪▪ Ethernet LAN
▪▪ RJ12 – Cash Drawer printer interface

20. Invoice:

1. The invoice is also part of homologation / fiscal certification process **only if the printing invoice process is included in the fiscal POS application**. If the printing invoice process is done by the other application, will be not a part of homologating process.
2. When a fiscal receipt is issued, it can be printed also an invoice, but only at the request of the customer.
3. According with the fiscal law it will be possible to use **simplified invoice** (fiscal ticket with CIF) or fiscal invoice.
4. The invoices can be managed by the till and generated by the till printer. They can also be printed from a separate back office program. This second solution is easier to support because, it doesn't require a homologation / fiscal certification by the Romanian fiscal authorities
5. The invoice can be generated on a laser printer or on the Fiscal Printer as a non-fiscal ticket. It is not required that the invoice to be signed or stamped by the customer and retailer.
6. According with the new regulation (new Fiscal Code) the **layout of the invoice is free to be chosen**. Only the mandatory fields will be necessary to be printed.
7. The software must manage a **range of serial and unique number of the invoice**.
8. All the invoices must be collected in the database of the software application in order to be used as a source for 394 Declaration (monthly declaration), which contains following information's: Costumer, Costumer Tax ID, Total VAT, Total sum.
9. According with the Romanian "Fiscal Code 2016" on the invoice it must be displayed the following **mandatory fields**:
    a. Number of the invoice (The invoice must have one prefix (Serie) and one unique number allocated by Retailer which afterwards must keep a record of this allocation).
    b. Date of invoice
    c. Name, address and CIF (fiscal ID, code de identification fiscal) of the seller
    d. Name, address and CIF (fiscal ID, code de identification fiscal) of the buyer
    e. Name of article
    f. Quantity of article
    g. Unit price (with VAT – netto price)
    h. VAT Rate and VAT Value of article
    i. Total VAT of invoice
    j. Total Value of invoice
10. Value of the Unit Price (without VAT – netto price) must be calculated from Brutto Sales Unit Price.
11. For the correct **calculation for the invoice**, it is necessary to use **4 decimals**

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12. Short invoice layout is allowed - on the invoice we can print only one sentence "Products according with fiscal receipt number *x* from date *y*". The invoice must be delivered together with the fiscal receipt.
13. Regarding the "Invoice printing" function, the retailer must have the possibility to issue the invoice for a fiscal receipt that was issued in the past – history of 50 days.
14. Invoice examples can be found in deliverables from RBC.
15. On the invoice some specific text messages can be printed. It is recommended to store them in the database and make them available to be selected for printing:
    a. Example: "Produsele nu sunt destinate comercializarii" (eng: Products are not for sales), "Produsele include taxa verde" (Products include Green Tax);

22. Homologation / fiscal certification:

1. A fiscal homologation / fiscal certification has to be performed by the Fiscal Authorities on the fiscal solution (POS application, fiscal printer, customer display). The institute is the National Institute for Research and Development in Informatics (ICI).
2. The homologation / fiscal certification will allow:
    a. The fiscal solution to be distributed and used in Romanian market
    b. The installation of the fiscal solution in retailer's stores.
    c. The maintenance of the fiscal solution
3. During the homologation / fiscal certification, a lot of tests will be performed in order to check that the fiscal solution respects the Romanian fiscal regulations, also the new introduced regulation regarding Electronic Journal and Data Transmission to Fiscal Authority.
4. The fiscal solution (hardware and software) must be sold by a Romanian company.
5. The retailer has to buy the fiscal solution from a fiscal distributor.
6. Once the homologation / fiscal certification is performed, the POS application can't be modified. The EJ fiscal law imposes a new technical certification in case of modification of POS application and fiscal firmware of the fiscal printer.
7. Steps of the homologation / fiscal certification:
    a. Provide a translated user guide in Romanian language
    b. Documentation for POS application in Romanian language – like user manual for cashier (including technical descriptions and interaction cash desk display + cash desk + Printer + receipt)
    c. All messages, buttons and screen on the POS application must be in Romanian language
    d. Provide a **hardware platform with the fiscal kit**, the till and the POS application
    e. Provide POS application name and version that needs to be certified.
    f. Provide fiscal documents related to the Fiscal controller (provided by the integrator)
    g. ICI (Institute) will test on their own:
        - all fiscal function
        - all EJ functions

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- all Data Transmission functions
- empty memory
- memory near full
- memory full
- empty EJ SD Card
- EJ SD Card near full
- EJ SD Card full
- Error handling in different areas of the receipts and reports
- Overflow tests
- Fiscal printer fiscalization tests from the Programming Mode of the POS application
- Issue reports from MF and EJ (printed and exported on the USB stick)
- For the initial technical certification there will be **NO tests** related to communication between the Fiscal Printer and the ANAF authority server.
    a. All tests will be done using Profile 0 (PreOperational).
    b. During the initial ICI certifications, there will be no ANAF test servers or simulator available for the tests.
- The tests related to communication between Fiscal Printer and ANAF servers will be done in the future extensions of the technical certifications.

8. The tests will perform electrical shutdown of the different peripherals in different states of an open fiscal ticket
9. Tests for missing paper, cover open of fiscal printer in different states of an open fiscal ticket
10. The test dates have to be planned at least 2 weeks in advance.
11. After Technical Homologation / fiscal certification, the fiscal authorization for delivery, installing and service will be obtained from the Ministry of Finance.
12. After fiscal authorization from Ministry of Finance, the fiscal solution can be sold and used in the store.

23. GET Commands:

1. GET commands output ASCII data strings.
2. GET commands never print anything.
3. MFC status or other CMOS data are never affected by GET commands.
4. All numbers are sent without leading zero.
5. Decimals are always implied for amount and percentage values (e. g.: "123" means "1,23")
6. GET commands are not executable if an error is pending
7. GET commands are accepted from display port or application port and the answer comes back on the same port.
8. Below there is the list of Get Commands available:

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a. Get commands related to receipts:
These commands can be used to gather information related to current receipt.
- GET_RECEIPT_TOTALIZER
- GET_RECEIPT_BRUTTO_SALES
- GET_RECEIPT_VAT_AMOUNT
- GET_RECEIPT_PAYMENT

b. Get commands related to fiscal day:
These commands can be used to gather information related to current fiscal day.
- GET_DAY_TOTALIZER
- GET_DAY_BRUTTO_SALES
- GET_DAY_VAT_AMOUNT
- GET_DAY_PAYMENT
- GET_DAY_STATISTIC
- GET_DAY_DRAWER_CONTENT
- GET_DAY_RECEIPT_TYPE_CNT GET_DAY_RECEIPT_GROUP_CNT
- GET_DAY_EVENT_CNT

c. Get commands related to Grand Totalizers:
These commands can be used to gather information related to Grand Totalizers.
- GET_GRAND_BRUTTO_SALES
- GET_GRAND_VAT_AMOUNT
- GET_GRAND_FISCAL_CNT

d. Get commands related to Fiscal Memory:
- GET_MFMEM_CNT
- GET_MFMEM_LAST_BL_DATE
- GET_MFMEM_LAST_BL_NUM
- GET_MFMEM_LAST_LOGICAL_BL_NUM
- GET_MFMEM_BL_NUM
- GET_MFMEM_BL_HEADER
- GET_DATE_TIME
- GET_VAT_RATES
- GET_CURRENCY_COUNTRY
- GET_TAX_TERMINAL_MODE
- GET_PAYMENT_LIST
- GET_ERROR_LIST
- GET_FIRMWARE_VERSION
- GET_TKD_INFO
- GET_HEADER
- GET_SNO_NUMBER
- GET_DATE_FIRST_RECEIPT
- GET_TAX_REPORT

9. EJ GET Commands

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These commands can be used by POS application in order to check the status and different parameters related to Electronic Journal. According to the status read the POS application can take appropriate action.

- EJ_GET_STATUS
- EJ_GET_SIZE
- EJ_GET_FREE_SPACE
- EJ_GET_INIT_INFO
- EJ_GET_MEDIUM_INFO
- EJ_GET_LAST_MEDIUM_INFO
- EJ_GET_EOD_ENTRY
- EJ_GET_NUMBER_OF_EOD
- EJ_GET_EOD_RANGE
- EJ_GET_DATE_RANGE
- EJ_GET_DATA
- EJ_GET_EOD_NUMBER_BY_DATE

10. MFC Commands

These commands can be used by POS application in order to manage the status of the fiscal memory and fiscal printer. According to the status read the POS application can take appropriate action.

- MFC_STATUS
- MFC_SYNC
- MFC_TEST
- MFC_INFO
- MFC_OPEN_D
- MFC_READ_LAST_INTERRUPTED_CMD
- MFC_READ_HOST_DATA
- GS_ENQ

24. Administrative Information:

1. It will be retailer's responsibility to store and archive:
    a. Z Report on paper for 5 years
    b. Fiscal memory for 10 years
    c. Electronic Journal Media (SD Card) for 5 years
2. There are two special books which belongs to each printer
    a. Book of service technician intervention (book for fiscal maintenance)
    b. Register (emergency book – manual sales written into the Register in case of failure to work of fiscal solution)
3. RBC can offer a solution to archive fiscal memory, EJ SD Cards and also the two paper books.
4. To perform some operations on fiscal printers, procedures have to be performed by sub-contractors approved by the Romanian fiscal authorities:
    a. importing from abroad fiscal products
    b. delivery on the Romanian market of fiscal products

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c. installation of the printer with initialization of the fiscal memory, electronic journal and Data Communication Module (data transmission module)

5. The printer can be moved from one location to another as long as the CIF number remains the same *and it's in the same county/district*. EJ printers have the possibility to change the address inside the fiscal header. No more than 16 fiscal header changes
6. During the fiscalization procedure the printer receives a fiscal ID, which is paired with the manufacturer serial number of the fiscal printer for the lifetime period of the device.
7. All the movements from one location to another of the fiscal devices starting with the import or production in Romania must be registered at the Fiscal Authorities in the Electronic Register of AMEF (fiscal devices).

25. Restrictions

1. **Tabacco *(Law no. 349/2002)***
   a. It is forbidden to sell tobacco products to persons under the age of 18.
   b. It is forbidden to sell cigarettes at piece.
   c. It is forbidden to sell cigarettes packs containing at less than 20 pieces.
   d. It is forbidden to sell tobacco products by automated means.
2. **Alcohol *(Law no.61/1991)***
   a. It is forbidden to serve alcoholic beverages, in public places, a drunk customer, as well as persons under the age of 18.
3. **Fireworks/Explosives *(Law no.126/1995)***
   a. The sale of pyrotechnic articles is only made by authorized legal persons and only in places authorized under the law.
   b. Individuals can't manufacture, own, sell, use or perform any other operation with pyrotechnic articles of categories F2-F4, T2 and P2.
   c. Pyrotechnic articles for entertainment from category F1 may only be made available on the market to persons who have reached the age of 16;
   d. Pyrotechnic articles for stage from category T1 and other pyrotechnic articles from category P1 may only be made available on the market to persons who have reached the age of 18.
4. **Weapons *(Law 295/2004)***
   a. In Romania, the weapons are classified according to three types:
   b. Prohibited weapons and ammunition – category A;
   c. Lethal weapons and ammunition subject to authorization – category B;
   d. Non-lethal weapons and ammunition – category C, D and E.
   e. The retail sale of weapons from category A is prohibited; they may be owned and used only by public institutions with responsibilities in the field of defense, public order and national security.
   f. Lethal weapons from category B can be owned or, as the case, worn and used by individuals only on the basis of weapon license, and by legal persons can be owned and used on the basis of the authorization.

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g. Non-Lethal weapons from category C and D can be owned and, as the case, worn and used by individuals or legal persons in subject to authorization (weapons classified in category C) or prior notice (weapons classified in category D).
h. Non-lethal weapons and device from category E can be owned and, as the case, worn and used by individuals and legal persons, without completing the registration or authorization procedures.

5. **Knifes, Blades *(Decree no. 163/1950)***

a. They are considered white weapons with a blade, weapons that meet the following criteria:
   - The blade is either solid with the handle or equipped with a system that allows to make a joint body with his handle;
   - Has a double edge throughout its length;
   - The length is greater than 15 cm;
   - Width is greater than or equal to 0,4 cm;
   - Has a handle provided with guard.
b. The port of white weapons, as well as their possession, is permitted only with authorization.

6. Pharmaceuticals

a. Retailers are allowed to sell only special categories:
   - non-prescription medicines;
   - self-medication medicines;
b. veterinary medicines that can be bought without a prescription.

26. Declaration 394 file structure

The 394-declaration file format currently accepted in Romania is:

The file must be text type and must always have the following name:
394_SSAA_JCCCCCCCCCC.TXT

EX:

394_S117_J11588780.txt

394 - fixed
SS - the semester (S1 for 01/01 to 30/06 and S2 for 01/07 to 31/12 periods)
AA - the last two digits for the fiscal year (e.g. 07 for 2007)
J - fixed
CCCCCCCCCC - fiscal code (registration code for VAT purposes)

The text file contains two types of records:

First record contains the identification data of the VAT payable company and the fiscal representative, reporting period and a totalizer of the declaration:

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No.	Field name/	Field type(Maximum length)	(Remarks)
1	Declaration code	N(3)	(394)
2	Fiscal code	N(10)	(Merchant fiscal code)
3	Semester	C(2)	(S1 or S2)
4	Year	N(4)	
5	Company name	C(100)	Merchant company name
6	Complete Address of the company	C(160)	Complete address of merchant
7	Phone	C(30)	
8	Fax	C(30)	
9	Email	C(80)	
10	Fiscal code of the fiscal representative	N(10)	
11	Fiscal representative name	C(100)	
12	Complete address of the fiscal representative	C(160)	
13	Phone of the fiscal representative	C(30)	
14	Fax of the fiscal representative	C(30)	
15	Email of the fiscal representative	C(80)	
16	Total number of suppliers/customers	N(15)	
17	Delivery base	N(15)	(Total amount of the sum of all netto totals on all issued invoices)
18	Delivery VAT	N(15)	(Total VAT for point 17)
19	Purchasing base	N(15)	(Total amount of the sum of all netto totals on all purchasing (received) invoices)
20	Purchasing VAT	N(15)	(Total VAT for point 17)

The next records will contain information regarding the goods deliveries/services and goods/services purchasing on national territory under the following format:

No.	Field name/	Field type(Maximum length)	(Remarks)
1	Operation type	C(1)	(A for purchasing or L for delivery)
2	Fiscal code of beneficiary/supplier/servicer (without leading RO)	N(10)	(Merchant fiscal code)
3	Name of beneficiary/supplier/servicer	C(100)	
4	Taxable base	N(15)	(sum of all invoice nettos)
5	VAT	N(15)	

Final remarks:

- Every record will end with "carriage return" and "linefeed".
- No Romanian special characters are accepted
- No control characters are accepted except those mentioned above
- Field separator is comma (,)
- Character type delimiter is #.
- The order of the fields is mandatory no matter if it's filled in or not!

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IMPORTANT NOTE:
Will be not in charge of retail software application to create the Declaration 394.
The software POS application must export to the software in charge to create the Declaration 394, the following fields:

Nr. Crt.	Romanian	English	Format	Obs.
1	"Nr.Crt."	Nr.Crt.	N(5)	
2	"Data",	Date	N(8)	
3	"Business date",	Business Date	N(8)	
4	"Nume magazin",	Store Name	C(64)	
5	"Serie factura",	Invoice serial	C(10)	
6	"Numar factura",	Invoice number	N(10)	
7	"Nume client",	Client Name	C(64)	
8	"Cod fiscal",	Fiscal Code	C(10)	
9	"Total fara TVA",	Delivery Base	N(15)	Delivery Base = Total Netto Sales on VAT A + B + C + D + E
10	"Valoare TVA",	Delivery Vat	N(15)	Delivery VAT = Total VAT on VAT A + B + C + D + E
11	"Total factura",	Invoice Total	N(15)	Invoice Total = Delivery Base + Delivery VAT
12	"Baza Cota A",	Total Netto Sales on VAT A	N(8)	
13	"Tva Cota A",	Total VAT on VAT A	N(8)	
14	"Baza Cota B",	Total Netto Sales on VAT B	N(8)	
15	"Tva Cota B",	Total VAT on VAT B	N(8)	
16	"Baza Cota C",	Total Netto Sales on VAT C	N(8)	
17	"Tva Cota C",	Total VAT on VAT C	N(8)	
18	"Baza Cota D",	Total Netto Sales on VAT D	N(8)	
19	"Tva Cota D",	Total VAT on VAT D	N(8)	
20	"Baza Cota E",	Total Netto Sales on VAT E	N(8)	
21	"Tva Cota E"	Total VAT on VAT E	N(8)	

27. EJ Reports

EJ reports must added to the POS application Report Menu.
In fact, will be two reports from EJ:

- *EJ_REPRINT_T2T_BY_SESSION*
- *EJ_REPRINT_T2T_BY_DATE_TIME*

For both reports will be necessary to choose one parameter related to:

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- Print report on the fiscal printer station – parameter = 2
- Download report to serial COM port – parameter =16

Before print/download the report, the cashier must choose the range of report:

- Select the Session to Session (Fiscal Day to Fiscal Day)
- Select the Ticket to Ticket

It is mandatory to check and validate the range, otherwise if the range is not correct the report will be not print/download or will be empty
In order to check the range of Session and Ticket can be used "EJ_GET_INFO_PER_SESSION". This command will return the range of the Ticket in the specific Session.

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